

Survey Paper

Approaches to the truck-drone routing problem: A systematic review

Jie Duan^{a,b,1}, He Luo^{a,b,c,1,*}, Guoqiang Wang^{a,b,c}^a School of Management, Hefei University of Technology, Hefei 230009, PR China^b Key Laboratory of Process Optimization and Intelligent Decision-Making, Ministry of Education, Hefei 230009, PR China^c Engineering Research Center for Intelligent Management of Aerospace System, Anhui Province, Hefei 230009, PR China

ARTICLE INFO

Keywords:

Literature review

Truck-drone routing problem

Vehicle routing problem

Approach

Drone

ABSTRACT

In recent years, research on collaborative routing between trucks and drones (also known as unmanned aerial vehicles, UAVs) has received considerable attention, with applications gradually emerging across various fields. This paper aims to summarize the theoretical research on the truck-drone routing problem from 2015 to 2024, categorizing, introducing, and comparing the approaches used, with a particular focus on their innovative aspects. It then collects public datasets and extension techniques for various types of problems. Typical application domains and the current state of the art in the field are then presented, laying the groundwork for scenario transition. Finally, based on the discussion, future development trends and research directions in the field are predicted.

1. Introduction

With the rapid growth of e-commerce and delivery demand, traditional logistics methods are challenged to handle complex urban traffic and remote deliveries. The truck-drone routing problem (TDRP) combines truck capacity with drone flexibility, enhancing efficiency and showing broad potential in logistics, inspection, and intelligence surveillance and reconnaissance (ISR). TDRP, a complex variant of the traveling salesman problem (TSP) and the vehicle routing problem (VRP), integrates coordination between trucks and drones by addressing drone route and truck routing constraints. Illustrative examples of TSP, VRP, and TDRP are shown in Fig. 1, where TDRP takes the collaboration of a truck and a drone as a case. The collaborative operation of trucks and drones, which emerged in the early 2010s to improve delivery efficiency in challenging urban environments, goes beyond traditional two-dimensional logistics by using drones to access hard-to-reach areas and trucks as bases, thus exploiting three-dimensional space to avoid congestion, optimize space utilization and reduce labor costs. The key motivation is to integrate practical applications with theoretical research, where practical applications inspire theory, and theory advances real-world applications. This paper categorizes and summarizes research methods in TDRP, aiming to outline theoretical approaches and promote future research for practical applications.

The rapid advancement of drone technology and its application in

delivery has further propelled the research and practical implementation of TDRP. Drone-based delivery services have evolved significantly since Amazon announced Prime Air in 2013, which aimed to deliver packages via drones within 30 minutes. Although Prime Air didn't become a reality until 2022 [1], other companies, like DHL, began experimenting with drone deliveries earlier, such as the Parcelcopter project in 2014 [2], which was later canceled in 2021 [3]. Meanwhile, drones are being explored for innovative applications in various sectors. Major companies such as Alibaba [4] and JD.com [5] have successfully tested drone delivery services in urban and rural areas, and Meituan is exploring drone-based food delivery [6]. The COVID-19 pandemic further accelerated the adoption of drones for contactless delivery of medical supplies, improving public acceptance [7]. The development of drone technology has expanded its range of applications and introduced it into the realm of TDRP, bringing new elements and mechanisms to the field. Enhanced drone capabilities, such as longer flight ranges and increased payload capacities, have further complicated the coordination between trucks and drones. In this context, there is an urgent need for efficient algorithms to optimize the problem in real time, addressing the uncertainties inherent in dynamic environments.

As theoretical research and practical applications continue to advance, numerous reviews have systematically summarized various aspects of TDRP. However, these reviews predominantly focus on a specific field or particular components of the problem. References

* Corresponding author at: Hefei University of Technology, NO. 193 Tunxi Road, Hefei, Anhui Province, PR China.

E-mail address: luohe@hfut.edu.cn (H. Luo).

¹ These authors contributed to the work equally and should be regarded as co-first authors.

[8–11] and [12–16] investigate variants of the relevant problems from the perspective of trucks and drones, respectively. Li [17] and Sluijk [18] examine the modeling approaches from a two-echelon perspective. Boccia et al. [19] discuss the classifications and methods of the flying sidekick TSP (FSTSP). Compared with the above, this paper systematically categorizes TDRP approaches, analyzes various application scenarios, and integrates theory with practice to offer a comprehensive reference framework for researchers and practitioners while suggesting future research directions to advance TDRP development.

Through a comprehensive search of databases including Web of Science, Science Direct, Springer, Taylor & Francis, Wiley, and INFORMS, this study synthesized research on the TDRP from 2015 to 2024. Prioritizing peer-reviewed journal articles, 1778 initial papers were identified using keywords such as "vehicle", "drone", "routing", "truck", "UAVs", "traveling salesman problem", and "vehicle routing problem". After merging and eliminating 604 duplicates, 1174 articles remained. Based on title and abstract relevance, 679 studies were excluded, leaving 495 for further review. The selection criteria were refined to focus on practical TDRP applications, the feasibility of solution approaches, and journal quality. Priority was given to studies with robust, implementable algorithms or optimization techniques, followed by those demonstrating real-world TDRP applications. Emphasizing high-quality, peer-reviewed journals ensured academic rigor, ultimately resulting in an in-depth analysis of 184 key publications.

This paper categorizes and reviews approaches to the TDRP using exact, heuristic, metaheuristic, approximation, and artificial intelligence (AI) algorithms. Exact algorithms provide optimal solutions for small-scale problems, while heuristic and metaheuristic algorithms offer efficiency and global optimization for more complex problems. Approximation algorithms deliver reasonable solutions within limited computational time, ideal for resource-constrained scenarios. AI algorithms enable intelligent task allocation and real-time optimization in dynamic environments. This classification systematically organizes TDRP research, offering insights into each algorithm's characteristics and guiding researchers and practitioners in selecting appropriate approaches.

The remainder of this article is structured as follows. Sections 2 to 6 systematically review the main categories of algorithms relevant to TDRP, including exact, heuristic, metaheuristic, approximation and AI algorithms. Section 7 presents common datasets and strategies for their improvements. Section 8 describes the main application fields and the current state of TDRP research. Section 9 provides the discussions including the approaches and applications covered in the previous sections. Future research directions are outlined from four perspectives in Section 10, followed by the conclusion in Section 11. The abbreviations used throughout this article are listed in the Appendix.

2. Exact algorithms

Exact algorithms offer a unique advantage in solving TDRP by guaranteeing the optimality of the solutions. These algorithms work by building a mathematical model of the problem, taking into account

various variables and constraints, which allows for the quick determination of optimal solutions for small-scale problems. Mathematical verification ensures the integrity and stability of the results, making these methods highly reliable. Common exact algorithms include the branch and bound (B&B), branch and cut (B&C), branch and price (B&P), dynamic programming (DP), column generation (CG), and bender decomposition (BD). These are versatile tools that can address a range of problems in linear programming (LP), integer programming (IP), and beyond.

However, the effectiveness of exact algorithms is subject to the scale of the problem at hand. While they perform exceptionally well for smaller problems, the computational time required grows exponentially as the problem size increases. This inherent limitation makes exact algorithms more suitable for tackling small-scale problems where their computational intensity is manageable and the need for guaranteed optimality is paramount. Consequently, despite their ability to provide accurate and stable solutions, the practical application of exact algorithms is often limited by the computational resources available and the scale of the problem to be addressed.

Related studies are listed in Table 1.

2.1. Branch and bound

By pruning and searching the problem space, branch and bound (B&B) can ensure that the final solution is the global optimal solution or an approximate solution of the optimal solution and is suitable for multiple combinatorial optimization problems. However, when the algorithm has high time complexity due to its use in large-scale problems, extending it to multi-objective problems becomes challenging. Therefore, continuous optimization and improvement can be achieved by expanding the search range and optimizing the branch strategy. Poikonen et al. [20,21] made each node of the search tree in the B&B correspond to a potential subsequence of the target access order, and the lower limit of a node was obtained by solving the second-order cone program. Dell'Amico et al. [22] used a depth-first approach to access the search tree nodes and expanded these nodes based on the order of their creation. Teimoury and Rashid [23] constructed new branches according to the heuristic customer selection rules.

2.2. Branch and cut

TDRP can be constructed as a series of complex mathematical models, and branch and cut (B&C) is an efficient algorithm for solving mixed integer programming (MIP) problems. In the BB framework, effective inequalities are introduced to cut the plane, which greatly reduces the number of branches and yields improved search efficiency. Nonetheless, a large number of subproblems must be solved, resulting in an exponentially increasing computational complexity and a long running time. The complexity and computational time can be improved based on the joint timing of the cutting plane, branch strategy, node selection, and so on.

Manyam et al. [28] integrated constraints and separation algorithms

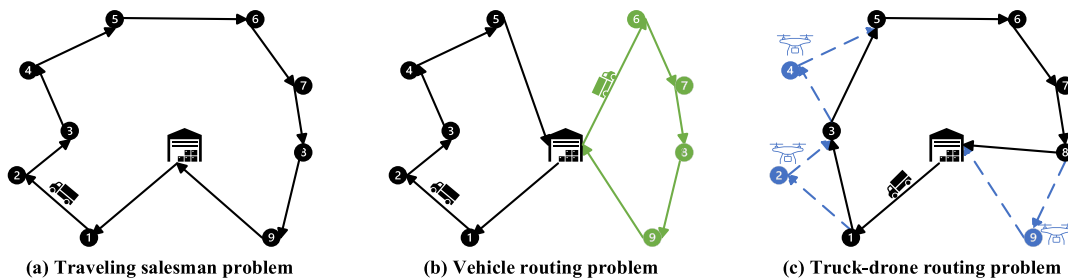


Fig. 1. Illustrative examples for the TSP, VRP, and TDRP.

Table 1
Literature overview of exact algorithms.

Alg	Reference	Appli	Pro	Mod	*T	*D	Obj	Truck visit	Drone visit number	Different L-R
B&B	[20]	Log	TSP-D	-	1	1	CT	Y	1	Y/N
B&B	[21]	ISR	MDRP	-	1	1	CT	N	1	Y
B&B	[22]	Log	FSTSP	-	1	1	CT	Y	1	Y
HB&B	[23]	Log	HTDRP	MILP	1	1	CT	Y	1	N
B&C	[24]	Log	FSTSP	MILP	1	1	CT	Y	1	Y
B&C	[25]	Log	FSTSP	MILP	1	1	CT	Y	1	Y
B&C	[26]	Log	TSP-D	MILP	1	1	CT	Y	1	Y/N
B&C	[27]	Log	mFSTSP	MILP	1	M	CT	Y	1	Y
B&C	[28]	ISR	CAGVRP	MILP	1	1	CC	Y	M	N
B&C	[29]	ISR	CAGVRP	MILP	1	1	CC	Y	M	N
B&C	[30]	Log	VRPD	MILP	M	N	CT	Y	1	Y
B&C	[31]	Log	IRP-D	MILP	1	1	CC	Y	1	Y
B&C	[32]	Log	PDSMTSP	MILP	M	M	CC	Y	1	N
B&C	[33]	Log	TSP-mD	MILP	1	M	CC	Y	1	Y/N
B&C	[34]	Log	FSTSP	MILP	1	1	CT	Y	1	Y
B&C	[35]	Log	OP-mD	MILP	1	M	Profit	Y	1	Y
B&C	[36]	Log	TDTL	MILP	1	1	CT	Y	1	Y
B&P	[37]	Log	VRPD	MIP	M	N	CC	Y	M	Y
B&P	[38]	Log	TSP-D	MILP	1	1	CT	Y	1	Y/N
B&P	[39]	Log	2E-VRP-D	MILP	M	N	CT	N	1	N
B&P	[40]	Log	RDTDP	-	1	1	Profit	Y	1	N
B&P	[41]	Log	BEVTSPD	MILP	1	1	CT	Y	1	Y
B&P&C	[42]	Log	VRPD	MIP	M	M	CC	Y	1	Y
B&P&C	[43]	Log	TDRP-TW	MILP	M	N	CC	Y	1	Y/N
B&P&C	[44]	Log	VRPLD	MIP	M	N	CC	Y	M	Y
B&P&C	[45]	Log	TD-DRPTW	MILP	M	M	CC	Y	M	Y
B&P&C	[46]	Log	TSP-D	DP	1	1	CT	Y	1	Y/N
B&P&C	[47]	Log	VRPSDP-TDH	MILP	M	N	CC	Y	1/M	N
CG	[48]	ISR	CDOP	MILP	1	1	Profit	N	M	Y/N
CG	[49]	Log	FS-TSP	ILP	1	1	CT	Y	1	Y
CG	[50]	Log	TDCDS	MILP	M	N	CC	Y	1	Y
CG	[51]	Log	TDPDP	MILP	M	N	CC	Y	M	Y
CG	[52]	Log	2EVRP-HD-UAV	MILP	M	N	CC	N	M	N
CCG	[53]	Pat	VRPDH	MILP	M		CC/CT	N	M	Y
DP	[54]	Log	TSP-D	IP	1	1	CC	Y	1	Y/N
DP	[55]	Log	TSP-D	-	1	1	CC	Y	1	Y
DP	[56]	Log	TSP-DR	MIP	1	1	CC	Y	1	N
BD	[57]	Log	HDTRP	MIP	1	M	CT	Y	1	N
BD	[58]	Log	TSP-D	MIP	1	1	CT	Y	1	Y/N
BD	[59]	Pat	VRPD	MIP	M	M	CC	Y	M	Y
BD	[60]	Log	VDCDRP	MICP	M	N	CC	Y	1	Y

Note: In the first row, "alg", "appli", "pro", "mod", "*T", "*D", and "obj" denote algorithm, application, problem, model, the number of the truck, the number of drones, and objective respectively. "Different L-R" means the launch and recovery nodes of the drone are different. A "-" in the table indicates that the content is not clearly marked in the paper. The tables in the following text are also abbreviated according to this rule.

into a B&C framework. And later, Manyam et al. [29] used LP and rounding-based heuristics to convert fractional solutions into feasible solutions, constraints were dynamically added to improve the convergence speed of the developed B&C. The incorporation of formulations with an exponential number of constraints into B&C solvers was achieved by Schermer et al. [26]. Dell'Amico et al. [24] introduced multiple enhanced constraint formulations in the way of cutting planes to separate feasible solutions from the search tree. Further formulations with a set of time variables were proposed by Dell'Amico et al. [25], which was more compact and efficient than previous constraints. Najy et al. [31] offered a number of extended effective inequalities and developed a B&C by initially relaxing and then adding individually lazy constraints. Dell'Amico et al. [27] proposed a 4-index model, which separates the imposed constraints by branching and cutting. Tamke and Buscher [30] introduced novel effective inequalities based on the characteristics of the given problem, established lower bounds for time variables, and suggested problem-specific cuts to strengthen linear relaxations. Saleu et al. [32] dynamically incorporated subtour elimination constraints into the B&C. Boccia et al. [34] presented novel and effective families of inequalities as well as different strategies for fixing variables. Morandi et al. [35] proposed inequalities for effectively breaking symmetry concerning an exponential number of constraints. Boccia et al. [36] enhanced B&C through two distinct series of valid inequalities. Tinic et al. [33] explored three B&C approaches.

2.3. Branch and price

Branch and price (B&P) is a combination of CG and B&B, resulting in high accuracy and low time complexity when addressing TDRP. While it requires computing upper and lower bounds and may limit the search space. Therefore, there is currently a strong research interest in developing reasonable acceleration strategies to improve the solution efficiency of the pricing problem within the framework of the B&P when solving TDRP. Roberti et al. [38] proposed several sets of effective inequalities to improve the continuous relaxation processes of mixed integer linear programming (MILP) models, applying three-level hierarchical branching and recursive DP methods to achieve improved computational efficiency. Wang and Sheu [37] converted the arc-based model to a path-based model and proposed an improved pulse algorithm to prune and expand the subproblem. Building on their work, Yang et al. [40] updated the Lagrange dual problem by subgradient search and applied the depth-first node selection strategy to rapidly solve the master problem. An improved impulse algorithm was proposed to prune and expand the subproblem, to quickly solve the master problem. Zhu et al. [41] employed a DP method to solve the subproblem within the B&B framework. Zhou et al. [39] designed a customized bidirectional labeling algorithm for the pricing problem.

B&P can be enhanced by adding effective inequalities to form the branch and price and cut (B&P&C), which improves its efficiency. Zhen

et al. [42] created a solution framework that incorporates CG, cut generator, and B&B, which utilizes efficient inequalities to enforce the linear relaxation of the model and employs a DP approach to resolve the pricing problem efficiently. Li et al. [43] employed CG to solve the linear relaxation of the original problem, added additional efficient inequalities to strengthen the lower bounds, obtained integer solutions through branching, and combined B&P&C with an adaptive large-scale search algorithm to solve large-scale problems. Xia et al. [44] created B&P&C using the Dantzig-Wolfe decomposition framework, improved the degradation and long-tail effect generated by the columns through Lagrange relaxation, and adopted a depth-first search strategy and a feasibility pump to obtain high-quality upper and lower bounds through. Yin et al. [45] embedded a variety of enhancement strategies into B&P&C. Tang and Hoeve [46] introduced two novel route relaxation methods based on decision diagrams.

2.4. Column generation

Column generation (CG) has high accuracy and low time complexity for addressing TDRP, and dynamically generates constraints during the solution process to obtain the optimal solution. It is well suited for large-scale LP. While constantly generating constraints and relying on LP solvers, the solution efficiency is limited. Hence, improvements can be made in terms of optimizing LP solvers, CG strategies, and incorporating mixed algorithms. Boccia et al. [49] developed two variable fixed strategies aimed at reducing the solution space, in addition to implementing five lower inequality bounds to strengthen constraints. These were then integrated into CG to enhance the algorithm's effectiveness. Zhang et al. [48] embedded acceleration methods such as tabu search (TS) and label loading into the CG framework to improve the iteration speed of subproblem. Gao et al. [50] proposed a variable neighborhood search (VNS) algorithm to solve the pricing problem. Gao et al. [51] integrated BD and DP into a CG framework. Fzia et al. [52] applied a CG approach to decompose a multi-phase problem into several pricing problems. Momeni et al. [53] employed a column and constraint generation approach for solving scenario-based stochastic robust optimization models.

2.5. Dynamic programming

Dynamic programming (DP) decomposes the given problem into several subproblems and solves them step-by-step, making it suitable for problems with overlapping subproblems and optimal substructure properties. DP solves the TDRP in phases, uses historical records to avoid repeated calculations, and achieves reduced time complexity. Nevertheless, it requires significant memory space, particularly when solving large-scale problems. To overcome this problem, the use of state compression to reduce the system's state space and parallel computing to speed up the solution process effectively improve the efficiency of the employed algorithm. Agatz et al. [54] employed DP to classify any given delivery sequence quickly. To reduce the running time of the algorithm, Bouman et al. [55] restricted the state set of each node. Dienstknecht et al. [56] transformed the problem-solving procedure into a two-phase solution and embedded different decomposition methods into polynomial-time DP to accelerate the algorithmic iteration process.

2.6. Benders decomposition

Benders decomposition (BD) is widely used to solve MIP problems. When solving a TDRP, the original problem is decomposed into a LP master problem and multiple subproblems by dividing constraints. This approach has several advantages, including low reliance on the LP solver, dynamic constraint generation, fast convergence speed, and the ability to handle multi-objective problems. The implementation of the algorithm is relatively complex, requiring effective division and a combination of constraints. To improve its efficiency, optimizing the BD

strategy is recommended. Vasquez et al. [58] developed an efficient BD for TSP with drone (TSP-D) by utilizing t-cut and t-reduction, which enhanced efficiency through effective inequalities and improved cutting strategies. Kang and Lee [57] introduced a logic-based BD for the discrete subproblem and used heuristics and variable bounds to accelerate the solution process. Momeni et al. [59] used the augmented epsilon constraint approach to solve the bi-objective MIP model and utilized the BD to address large-scale instances. Yin et al. [60] integrated BD and B&P, employing B&P to solve BD subproblems and introducing effective cuts in the master problem.

3. Heuristic algorithms

Heuristic algorithms play a crucial role in solving TDRP, especially when dealing with large-scale scenarios and various practical constraints. These algorithms, which are grounded in experience-based solution methods, are designed to find approximate solutions by fully considering the objectives, constraints, and the search space of a given problem. Although they do not guarantee the discovery of the optimal solution, heuristic algorithms are known for achieving good results in practical applications. This is largely due to their ability to adjust the search path based on either individual or global experience, thereby enhancing search efficiency.

Heuristic algorithms can be broadly classified into two categories: traditional heuristics and metaheuristics. Traditional heuristics include constructive heuristics, local search, and multi-phase heuristics. These categories encompass a variety of approaches aimed at constructing a solution piece by piece, refining solutions through local modifications, or employing a sequence of different strategies to explore the solution space more effectively. Table 2 presents the related research related to these methods. Despite their practical utility, heuristic algorithms have limitations such as the inability to guarantee the accuracy of results, potential convergence to local optima, and susceptibility to overfitting. These challenges highlight the need for continuous improvement in the algorithm's greedy strategy, search strategy, heuristic information used, and the application of hybrid methods. By addressing these challenges, the performance of heuristic algorithms in navigating the complex landscape of TDRP can be significantly enhanced, making them even more valuable for tackling the intricacies of large-scale logistical challenges.

3.1. Constructive heuristic

A constructive heuristic (CH) algorithm constructs an initial solution through rules or experience and uses the search algorithm for iterative optimization. The solution quality of the algorithm can be improved by introducing efficient heuristic functions and designing more scientific and reasonable construction rules. Murray and Chu [61] designed a CH algorithm based on reassignment and splitting. Mulumba and Diabat [65] optimized global routing by applying dynamic routing and reassignment strategies. Jeong et al. [62] used a combination of the CH and search heuristic to enhance results by generating initial solutions and performing multiple search operations with constructive rules. The search operators include 2-opt, swap, and insertion. Kundu et al. [63] developed a splitting algorithm based on the shortest path problem to process the initial solution and used a 2-opt local search to iteratively improve the results. The initial solution was generated by the Lin-Kernighan-Helsgaun approximation algorithm. Salama and Srinivas [64] integrated a knowledge-based constraint, an iterative K-means clustering algorithm, an acceleration strategy based on a warm-start technique, and a machine learning-based heuristic method. Es Yurek and Ozmutlu [66] reduced the number of parallel nodes and increased the number of serial nodes through heuristic methods to reduce the computational space of the problem. Liu et al. [67] designed two CHs based on the strategy of "cluster first, route second" (CFRS) and "route first, split second" (RFSS), and applied a local search strategy to improve

Table 2
Literature overview of heuristic algorithms.

Alg	Reference	Appli	Pro	Mod	*T	*D	Obj	Truck visit	Drone visit number	Different L-R
CH	[61]	Log	FSTSP	MILP	1	1	CT	Y	1	Y
TPCSA	[62]	Log	PDSTSP	MILP	1	1	CT	Y	1	Y
CH	[63]	Log	FSTSP	MILP	1	1	CT	Y	1	Y
CH	[64]	Log	JOCR	MINLP	1	M	CT/CC	Y	1	N
CH	[65]	Log	DAPDP	MILP	M	M	CC	Y	1	Y
HH	[66]	Log	TSP-D	MILP	1	1	CT	Y	1	Y
CFRS/RFSS	[67]	Ins	2L-PA-RP	LP	1	1	CT	N	M	Y
CH	[68]	ISR	2E-GU-RP	MIP	1	1	CT	N	M	Y/N
HCWH	[69]	Log	HVDRP	MIP	1	M	CC	N	M	N
RFCS	[54]	Log	TSP-D	IP	1	1	CT	Y	1	Y/N
HH	[70]	Log	PDSTSP	MILP	1	M	CT	Y	1	N
HH	[71]	Log	EM-TSPD	MILP	1	M	CT	Y	M	Y
TPH/SPH	[72]	Log	VRPD	-	M	N	CT	Y	1	Y
TPH	[73]	Log	VRPTW-D	MILP	M	N	CC	Y	1	Y
Decomposition	[74]	Log	TSP-D	MILP	1	1	CT	Y	1	Y
ODEA-ARA	[75]	Log	MCRP-DP	MILP	1	M	CC/CT	Y	M	Y
Heuristic	[76]	Log	CCHDP	MIP	1	1	CT	N	M	Y/N
RF-RRO	[77]	Log	CTDRSP-ELR	MILP	1	M	CT	Y	1	Y/N
TPH	[78]	Log	VRP-DTW	MILP	M	N	CC	Y	1	Y
TPH	[79]	Pat	TPRP-D	MILP	1	1	CT	Y	M	Y
TPH	[80]	Log	VRPD	MILP	1	M	CT	N	M	N
TPH	[81]	Log	MDVRPD-H	MILP	M	M	CT	Y	1	Y
IO-3S	[82]	Log	HDDP	IP	1	M	CT/CC	N	M	Y
MPH	[83]	Log	k-MDVRP	MILP	1	M	CT	N	M	Y/N
MPH	[84]	Log	mFSTSP	MILP	1	M	CT	Y	1	Y
MPH	[85]	Log	TSP-D/OTMD	LP	1	M	CT/*D	Y	1	Y
MPH	[86]	Log	mFSTSP-VDS	LP	1	M	CT	Y	1	Y
MPH	[87]	Log	TSP-SBO	MIP	1	1	CT	Y	1	Y
MPH	[88]	Log	-	MILP	1	M	Makespan	N	M	Y/N
MPH	[89]	Log	HSNTSP-D	MIP	M	N	CC	N	M	N

the solution obtained by the heuristic. Luo et al. [68] developed two constructive heuristic algorithms: one performs construction first and then splits, and the other performs construction first and then assigns. Karak et al. [69] extended the algorithm proposed by Clarke and Wright [90], generating a hybrid algorithm and designing three heuristic algorithms for the construction of an iterative framework.

3.2. Local search algorithm

The local search (LS) algorithm starts from an initial solution and performs optimization by continuously selecting better solutions in the neighborhood. Diversity strategies, randomness, and more efficient neighborhood structure designs can be introduced to optimize the algorithm to improve its convergence speed and solution accuracy. Agatz et al. [54] combined an LS and DP based on Beasley's [91] "route-first, cluster-second" idea. Dell'Amico et al. [70] used the fast heuristic method to generate initial solutions and then iterated through an LS process with random restarts. Mahmoudi et al. [71] embedded the LS into the CH and used the 2-opt search operator.

3.3. Multi-phase heuristic

The multi-phase heuristic (MPH) divides the given problem into multiple phases for solution, and adopts a variety of heuristic algorithms to achieve the effect of collaborative optimization in different phases. By introducing a more flexible phase segmentation strategy and designing a more effective combination method, as well as other methods, the search efficiency of the algorithm can be improved.

Schermer et al. [72] designed a two-phase heuristic (TPH) and a single-phase heuristic (SPH) based on the improved method and used the nearest neighbor (NN) heuristic to generate an initial solution. The TPH was constructed by 2-opt and string mixing, and the SPH was constructed with a combination of 2-opt, string mixing, drone insertion, and delivery exchange. Es Yurek and Ozmutlu [74] developed a heuristic iterative algorithm to decompose the problem to be solved into two

phases. The values of the upper-level decision variables are determined in the first phase, and the upper-level decision variables are fixed to optimize the remaining lower-level subproblems in the second phase. Di Puglia Pugliese et al. [78] embedded a two-phase strategy into a multi-start framework to form a TPH algorithm. Luo et al. [79] designed an improved TPH, first generating an initial solution and then extracting and assigning the solution. Shi et al. [81] introduced independent helicopter transportation based on the TDRP, a TS algorithm determines the routes for trucks and helicopters in the first stage, and nodes are assigned incorporating drone flights in the second stage. Wen and Wu [82] designed a new three-phase iterative optimization algorithm based on the divide-and-conquer strategy. In the first phase, clusters are divided. Then, variable neighborhood descent is combined with TS to generate and improve the initial solution during the second phase. Finally, in the third phase, DP is adopted to solve the given problems, and simulated annealing is introduced to iteratively optimize the three phases. Poikonen et al. [83] introduced a route-transform-shortest path (RTSP) three-phase heuristic algorithm. Murray and Raj [84] proposed a three-phase iterative heuristic algorithm. Li et al. [85] constructed a multi-phase iterative heuristic algorithm to continuously iterate over feasible sets to obtain feasible solutions. Raj et al. [86] used an LS to improve the quality of the obtained solution based on a three-phase heuristic algorithm. Gao et al. [89] developed a three-stage decomposition algorithm, combining VNS and the NN algorithm. Gu et al. [88] used ant colony optimization to determine truck stops, a hybrid genetic algorithm for drone paths, and local optimization to adjust overall routes.

4. Metaheuristic algorithms

Metaheuristic algorithms have emerged as highly effective strategies for tackling TDRP, benefiting from their ability to navigate complex search spaces without being tailor-made solutions for specific problems. These algorithms are designed with general search requirements in mind, providing advantages such as rapid execution, extensibility, and

robustness to variability in problem parameters. Notably, many metaheuristic algorithms draw inspiration from natural phenomena and processes, incorporating elements of randomness to enhance their search capabilities.

A major focus of current research on metaheuristic algorithms is the development of mechanisms that prevent premature convergence to local optima, ensuring a more exhaustive exploration of the solution space. This is achieved by various innovative algorithms such as genetic algorithm (GA), simulated annealing (SA), particle swarm optimization (PSO), ant colony optimization (ACO), tabu search (TS), iterated local search (ILS), variable neighborhood search (VNS), and adaptive large neighborhood search (ALNS), among others. These methods are characterized by their use of random interference and other strategies to avoid the common trap of heuristic approaches, such as settling for suboptimal solutions. Related research on metaheuristic algorithms is presented in Table 3.

The growing body of metaheuristic algorithms for TDRP underscores their versatility, effectiveness, and the potential for synergistic combinations. Compared to traditional heuristic algorithms, metaheuristics offer an enhanced ability to quickly find optimal or near-optimal solutions, often with reduced solution times. This advantage is due to their sophisticated algorithms that dynamically adapt their search strategies to bypass local optima and explore a wider range of potential solutions. As such, metaheuristic algorithms are a keystone of modern approaches to solving complex logistics challenges, offering a hybrid of speed, flexibility, and solution quality that is invaluable in the field of TDRP.

4.1. Genetic algorithm

Genetic algorithm (GA), as a swarm-based randomized search method, has been widely used in the research of TDRP. It has good adaptability and is suitable for optimization problems with multiple constraints. There are several weaknesses during the solution process, such as higher sensitivity to initial parameter settings, slow convergence speed, poor local search ability, a large number of control variables, and a lack of definite termination criteria. Given these challenges, the most direct strategy is to optimize the structure of the GA, which can be improved mainly based on the utilized coding method, initial populations, selection operator, crossover operator, mutation operator, parameter selection, etc.

Initially, improvements in the encoding method are explored. Binary coding exhibits high population diversity and high stability, but it requires large storage space, performs decoding, and is difficult to understand. Improved coding approaches often use decimal coding to control the model parameters to reduce the combination explosion and premature convergence problems. Real coding is used for a genetic search in a larger space to increase the accuracy of the algorithm [95]. Multi-layer coding is used to accurately map the solution space to the coding space to reduce the imposed genetic difficulty. Wang et al. [97] proposed an odd-even layered genetic algorithm (O-ELGA) for the MIP model, which uses parity coding to encode chromosomes and levels the chromosomes before performing selection, crossover, and mutation operations. Sitek et al. [98] developed an efficient dedicated GA using dedicated coding to effectively reduce the number of unsatisfied constraints. Mara et al. [105] created an innovative coding scheme by merging LS operators.

Improvements achieved with the selection, crossover, and mutation operators are then studied. The selection operator incorporates a fitness function into the algorithmic iteration process to evaluate the strengths and weaknesses of individuals. The focus of improvement is to select individuals with greater fitness and retain them for the next generation to reduce the selection error incurred. Commonly used crossover operators include single-point crossover, multi-point crossover, and uniform crossover operations. Improvements to these operators focus on improving their ability to identify optimal gene sequences. Mutation operators are applied to the offspring generated by the selection and

crossover operations. Simple mutation and uniform mutation are the most commonly used mutation operators. Their improved versions focus on improving the local search ability of the algorithm to prevent premature phenomena, such as adaptive mutation probabilities. Both Chiang et al. [93] and Li et al. [94] used two-point crossover and real-valued mutation operators in their GA designs. Mourelo et al. [92] used the K-means algorithm for clustering, used a GA to solve the given problem, and continuously updated the population through the random changes imposed on five mutations. Huang et al. [95] first set the maximum upper bound of the K-means clustering algorithm, real coding and a 2-opt crossover operator are used in the resulting GA. Karaköse [96] employed dynamic mutation to avoid getting stuck in local optima.

Other algorithms can also be combined within the framework of GA. Different bionic optimization algorithms have unique advantages, and they can be integrated to learn from each other's strengths. Wu et al. [104] incorporated the distribution estimation algorithm with GA, using random sampling on the established probability model to generate new solutions and adaptively adjusting the global and local search capabilities at different phases of the iterative process. Another possible approach is to integrate the GA with heuristic algorithms and search strategies. Ha et al. [99] proposed a new convergence-promoting recovery method and an adaptive penalty mechanism based on a splitting algorithm, a problem-specific intersection method, and a LS operator for dynamic population management and adaptive diversity control. Peng et al. [100] designed a crossover algorithm with a low visit cost to generate sub-solutions and used an LS to optimize the sub-solutions of the GA. Peng et al. [101] proposed a hybrid genetic algorithm (HGA) that includes a crossover algorithm with the minimum visit cost, a population management procedure, and a three-level hierarchical search algorithm. Lin et al. [103] decomposed the problem, solved the main problem with the GA, and solved the subproblems with a heuristic method. Euchi and Sadok [102] developed a mixed algorithm of sweep and GA, which enriched the diversity of the initial population by adopting the NN heuristic procedure and savings algorithm. The sweep algorithm was used to improve local search efficiency.

In addition, the nondominated sorting genetic algorithm (NSGA-II) with an elite strategy can be improved to solve multi-objective problems. NSGA-II introduces a fast nondominated sorting method, elite strategy, crowding degree, and a crowding degree comparison operator based on the NSGA, which reduces the computational complexity of the algorithm, maintaining the diversity of different groups and improving the algorithm in terms of its solution speed and quality. Wang et al. [106] redesigned the main components of NSGA-II, introducing a labeling algorithm to rank nodes and using the solution that provided the smallest ranks for chromosomes to calculate the crowding distance. Kuo et al. [107] improved the operators of NSGA-II by considering time and distance-dependent chromosomes, with a two-point crossover as the crossover operator and the mutation operator randomly selected among exchange, insertion, and inversion mutation. Luo et al. [108] constructed a hybrid multi-objective optimization algorithm according to greedy and Pareto LS. Based on the NSGA-II population framework, the greedy method was used to create an initial solution, and improved Pareto LS, global search, and genetic operations were introduced. Lu et al. [109] integrated a multi-objective evolutionary algorithm with LS to devise a variable-length two-dimensional array coding method, fitness evaluation criteria, multimodal mutation operation, and specialized LS operators. Yılmaz et al. [110] introduced three selection mechanisms, greedy, random, and fitness distance balance, into the evolutionary algorithm, effectively preventing the algorithms from converging to local optima.

4.2. Simulated annealing

Simulated annealing (SA) is a global statistical optimization method that has the advantages of accepting inferior solutions to a limited extent, jumping out of locally optimal solutions, and being suitable for

Table 3

Literature overview of metaheuristic algorithms.

Alg	Reference	Appli	Pro	Mod	*T	*D	Obj	Truck visit	Drone visit number	Different L-R
GA	[92]	Log	-	-	1	M	CT	Y	1	Y
GA	[93]	Log	GVRP-UAV	MILP	M	M	CE; CC	Y	1	Y
GA	[94]	Log	GVRP-UAV	MILP	M	M	CC; EC	Y	1	Y
GA	[95]	Ins	UAV-OV	MIP	1	M	CT; CD	N	M	N
GA	[96]	Log	HTMDRP	-	1	M	CT	Y	1	Y
O-ELGA	[97]	Ins	TTI-TSP-D	MIP	1	1	CT	N	M	Y
DGA	[98]	Log	EVRPD	BIP	M	M	CC	N	1	Y/N
HGA	[99]	Log	TSP-D	-	1	1	CT; CC	Y	1	Y
HGA	[100]	Log	-	MIP	1	M	CT	N	M	N
HGA	[101]	Ins	-	LP	1	M	CT	N	M	N
HGA	[102]	Log	VRPD	MILP	M	M	CT	Y	1	Y
h-GA	[103]	Log	-	MIP	1	M	CT	N	1	Y
h-PSO										
EDA-GA	[104]	ISR	-	MIP	M	M	CT	Y	M	Y
MA	[105]	Log	EVDRP	MILP	M	M	CT	Y	1	Y
INSGA-II	[106]	Log	TSP-D	MILP	1	1	CT; CC	Y	1	Y
NSGA-II	[107]	Log	VRPD	MILP	M	M	Makespan	Y	1	Y
HMOA	[108]	Log	Mo-CRPTW-mD	MILP	1	M	CC	Y	1	Y
HMOEA	[109]	Log	HVRP-SD-TVW	MILP	M	M	CT	Y	1/M	Y
FDB-EA	[110]	Log	TSP-D	MILP	1	1	CT	Y	1	Y
SA	[111]	Log	MDVRPSDPNF	MILP	M	M	CC	Y	1	Y
ATSA	[112]	Log	MTRPD	MILP	1	M	WT	Y	1	N
BOA-KS	[113]	Log	TmDTL	-	1	M	CC	Y	M	Y
TDRA	[114]	Log	STRPD	MILP	1	M	WT	Y	1	Y
TDRA	[115]	Log	FTRPD	MILP	1	M	CD	Y	1	Y/N
HH	[116]	Log	2E-RP-T&D	MILP	1	1	CC	Y	M	Y
OTS	[117]	Log	CTDRSP-FL	MILP	1	M	CT	Y	1	Y
ASALN	[118]	ISR	-	MILP	1	M	CC	Y	M	Y
AMS-SA	[119]	Log	VRP-D-MC	LP	M	M	Profit	Y	M	Y
MP-ISA	[120]	Log	MDRP-PD	MILP	M	M	CC	Y	M	Y
SA-TS	[121]	Pat	FPVRP	MILP	M	N	Profit	N	M	Y
ACO	[122]	ISR	-	-	1	1	CC	Y	1	Y
ACO	[123]	Log	-	MILP	M	M	TN; CT	N	1	N
CP-ACO	[124]	Log	-	MIP	M	M	CC; CS	Y	1	Y
ACO	[125]	Log	VRPD	MIP	M	M	CC	Y	1/M	Y
ACO	[126]	Log	EVRPD	MILP	M	N	EC	N	1	N
ABC	[127]	Log	MTSP-MHD	MILP	1	M	CC	Y	M	Y
MSTS	[128]	Log	mTSP-mD	MILP	1	M	Makespan	Y	1	Y
VNTS	[129]	Log	-	MINLP	M	M	CT	Y	1	Y
LS	[130]	Log	mFSTSP	MILP	1	M	CT	Y	1	Y
ILS	[131]	Oth	CVTSP	MISOC	1	1	CT	Y	M	Y
ILS	[132]	Log	TSP-MVD	MILP	1	1	CT	Y	M	Y
ILS	[133]	Log	OPDP-MTMV	LP	M	M	CC	Y	M	Y
ILS	[32]	Log	PDSMTSP	MILP	M	N	CT	Y	1	N
ILS	[134]	Log	TDHRP-TDPTT	MILP	M	M	CC	Y	1	Y
VNS-TS	[135]	Log	VRPDERO	MILP	M	M	Makespan	Y	1	Y
VNS	[136]	Log	VRPTWD	MIP	M	M	CC	Y	1	Y
VNS	[137]	Log	FSTSP-DR-MP	MILP	1	1	CT	Y	1	Y
VNS	[138]	Log	TSP-D	MIP	1	1	CT	Y	1	Y/N
HGVNS	[139]	Log	FSTSP	MIP	1	1	CT	Y	1	Y
GVNS	[140]	Log	mTSP-D	MILP	M	M	CT	Y	1	Y
GVNS	[141]	Log	VRPD-RS	MILP	M	N	Makespan	Y	1	Y/N
SINS	[142]	Log	MVTR-RP	MILP	M	N	CC	N	1	N
RVND	[143]	Log	FSTSP	-	1	1	CT	Y	1	Y
ILS-VND	[144]	Log	VRPD-MV	MILP	M	M	CC	Y	M	Y
RVND-SA	[145]	Pat	GVD-ARP	MIP	1	M	CT	N	M	Y
VND	[146]	Log	ATSP-D	MILP	1	1	CT	Y	1	Y/N
MTS			VS-ATSP-D							
ALNS	[147]	Log	TSP-mD	-	1	M	CC	Y	1	Y
ALNS	[148]	Log	VRPD	MIP	M	M	CC	Y	1	Y
ALNS	[149]	Log	MC-VRPTW-D	MILP	M	M	CC	Y	1	Y/N
ALNS	[150]	Log	FSTSP	MIP	1	1	CT	Y	M	Y
ALNS	[151]	Log	RTSP-mD	SOC	1	M	CT	Y	1	Y
ALNS	[152]	Log	TDRP-SA	MINLP	M	N	CC	Y	M	Y
ALNS	[153]	Log	MFS-VRPD	MILP	M	N	CC	Y	M	Y
ALNS	[154]	Oth	2L3P-MAC-ARP	MIP	1	1	CT	N	M	Y
ALNS	[155]	Log	2E-LRPD	MILP	M	N	CC	Y	1	Y/N
ALNS	[156]	Log	SARP-D	MINLP	M	M	CC	Y	1	Y
ALNS	[157]	Log	DAPDP	MILP	M	M	CC	Y	1	Y
ALNS	[158]	Log	mDPD-T	MIBP	M	M	CC	Y	M	Y

(continued on next page)

Table 3 (continued)

Alg	Reference	Appli	Pro	Mod	*T	*D	Obj	Truck visit	Drone visit number	Different L-R
IALNS	[159]	Pat	VD-ARP	MILP	1	M	CT	Y	M	Y/N
IALNS	[160]	Log	TDPDP-TW	MILP	M	M	CC	Y	1	Y
IALNS	[161]	Log	GVRPD-SR	MIP	M	M	CE	Y	1	Y
ALNS-SP	[162]	ISR	VRP-TDA	CP	M	M	CC	Y	1	Y
ALNS-SP	[163]	ISR	TDS	CP	M	N	CC	Y	1	Y
MS-IALNS	[164]	Log	MFDP-TTD	MILP	M	N	CT	Y	M	Y/N
LNS	[165]	Log	PDSTSP	MILP	1	M	CT	Y	1	N
GRASP	[166]	Log	TSP-D	MILP	1	1	CC	Y	1	Y
GRASP	[167]	Log	TSP-D	IP	1	1	CC	Y	1	Y
GRASP	[168]	Log	TSP-D	-	1	1	CT	Y	1	Y
GRASP	[169]	Log	2ER-TD	MILP	1	M	CT	N	1	N

obtaining global optimal or approximate global optimal solutions to optimization problems. SA is one of the most efficient algorithms for TDRP. Nonetheless, with the increasing amount of data processing and result refinement requirements, when many variables are given and the objective function is complex, the iterative operation speed of the algorithm is slow, many types of disturbance mechanisms are encountered, and the model flexibility is high. During the search process, the previous optimal solution is lost due to the high execution probability of the acceptance link. In response to the above problems, the inner loop, perturbation mechanism, and memory function of the algorithm can be improved.

To begin with, the search stability of the inner loop has been improved. The metaheuristic method based on a single solution needs to run the Metropolis algorithm many times to ensure its global search ability. However, running Metropolis too many times will waste time, while running it too few times may compromise the quality of the solution obtained. Next, a variety of random disturbance mechanisms have been utilized, as solutions are unlikely to be improved by a single perturbation mechanism. Afterward, the memory function has been enhanced. By remembering the current optimal solution throughout the search process, the search efficiency of the algorithm can be improved. This can be done by introducing a tabu list to archive forbidden actions.

In addition, an improved SA algorithm can be consolidated with other metaheuristic algorithms to form an efficient algorithm that achieves complementary functions and improved performance. Liu et al. [111] designed a two-stage heuristic algorithm based on SA, and proposed an acceleration technique. Moshref-Javadi et al. [112] combined SA and TS by using SA to control the main loop of the algorithm and probabilistically accept non-tabu and non-improved solutions, while using TS to avoid revisiting solutions during the iterative process. Liu et al. [116] proposed a hybrid heuristic algorithm based on SA in combination with the TS algorithm, added the infeasible solutions obtained by SA to the tabu list, and integrated the NN and cost savings strategies. Manshadian et al. [121] improved the initial solution with a customized SA-TS algorithm. Moshref-Javadi et al. [114] incorporated SA into the ALNS method. Gonzalez-R et al. [113] integrated SA with an iterative greedy algorithm, using a Kalman filter indicator as a stopping criterion. Tian et al. [118] developed an adaptive SA algorithm with a large neighborhood by utilizing SA as an iterative optimization framework and employing the nearest neighbor cost-savings strategy to obtain the initial solution. Moshref-Javadi et al. [115] combined various algorithms based on the work of [112,114] and used SA to obtain the initial solution, the elliptic customer assignment algorithm to obtain the truck path, and several heuristic methods to achieve subsequent improvements. Salama and Srinivas [117] constructed an optimization-capable two-phase search algorithm. In the first phase, SA is used to obtain a good feasible solution under limited area constraints, and in the second phase, a VNS is used to improve the solution. Amine et al. [119] proposed an adaptive multi-start SA algorithm that combines the SA framework with GA, a multi-start approach, ALNS, VNS, and other algorithms. The neighborhood of candidate solutions is explored iteratively to improve the diversity and searchability, and to avoid local

optima.

4.3. Ant colony optimization

By transforming the TDRP into an ant colony behavior problem, routing problems can be better solved. This application is relatively new and has been improved with further research. Ant colony optimization algorithm (ACO) is essentially a heuristic global optimization algorithm in the category of evolutionary algorithms, and it mainly consists of initialization, solution construction, and pheromone updating processes when used for TDRP. Given the shortcomings of ACO, such as slow convergence speed and easy stagnation, it can be improved by providing enhancements for the pheromone adjustment mechanism, search strategy, and method of integration with other optimization algorithms.

In the first place, the pheromone adjustment strategy has been improved. The process of the ACO converging to the optimal solution is a dynamic process of realizing pheromone positive feedback, and the selected pheromone adjustment strategy has a great level of influence on the convergence and solution efficiency of the resulting algorithm. Gu et al. [123] introduced variable visibility and multi-level feedback pheromones to obtain feasible solutions through the continuous three-level optimization and backpropagation of pheromones. Kyriakakis et al. [126] adopted pheromone update rules, local updates, and global updates for an ant colony system (ACS) variant and restricted the pheromone levels of max-min ant system (MMAS) variants to a certain interval to prevent premature convergence. Second, the effect of the search strategy on the performance of the algorithm is explored. Improving the search strategy of an algorithm is beneficial for increasing the path selection diversity and global search ability of the algorithm. Huang et al. [125] improved the ACS and adopted 2-opt as the LS.

Besides, the performance of the used algorithm can be enhanced by fusion with other algorithms. ACO has a strong positive feedback capability, and it can quickly converge to the optimal solution through pheromone accumulation and updating. In spite of that, the initial pheromone is insufficient, and the solution speed is slow. In consequence, good combinations of the ACO and other optimization algorithms can be used. Liang et al. [122] designed a path intersection operation in the GA for ACO to increase the diversity of the obtained solutions, avoid local minima, and accelerate algorithmic convergence. Kyriakakis et al. [126] introduced VNS, variable neighborhood descent, and other local parts based on the ACO framework. The search method limits the search within the feasible solution space and provides better solutions. Das et al. [124] incorporated the Pareto method into the ACO framework to solve multi-objective problems, giving a single objective different pheromones and random weights. And NSGA-II was used for verification. Luo et al. [127] integrated SA, TS, and elite selection strategies into the artificial bee colony framework, employing a specific neighborhood algorithm to explore new solutions.

4.4. Tabu search

Tabu search (TS) adopts neighborhood optimization, which is a

greedy neighborhood search algorithm that is suitable for discrete optimization while easily falling into local optima. For further application of TS to TDRP, improvements can be made in terms of the initial solution performance, iterative search process, and diversity search strategies, which are usually achieved in combination with other optimization algorithms.

First and foremost, the performance of the initial solution can be improved. TS is highly dependent on the initial solution. A good initial solution can make the TS process obtain a great solution in the solution space, while a poor initial solution reduces the convergence speed of TS. Luo et al. [128] embedded TS with customized neighborhood structures into a multi-start TS algorithm and improved the initial solution quality through a random sequence-based construction algorithm. Tailored relocation, swap neighborhood structures, and a two-stage feasibility evaluation method were adopted to expand the solution scale.

Secondarily, the search process of TS needs to be optimized. The iterative search process is a serial method involving only the movement of a single state, rather than a parallel search. By implementing parallel strategies in the initialization and parameter settings, various types of parallel TS algorithms can be obtained. The centralized search strategy is used to find the global optimal solution, and the diversity-based search strategy is used to widen the search domain. However, in the case where both concentration and diversity search are used, the achieved diversity is insufficient. The diversity can be increased by random reinitialization. Tong et al. [129] applied point exchange and link exchange to change the neighborhood structure for expanding the TS area.

4.5. Iterative local search

During the process of solving TDRP, iterative local search (ILS) introduces randomness by perturbing the current optimal solution and performs an LS in the neighborhood of the solution, which can allow it to avoid falling into local optimal solutions to some extent while gradually approaching the global optimal solution. To improve the global convergence and flexibility of the ILS algorithm, improvements can be made in terms of the perturbation method utilized, search strategy, and neighborhood structure.

The use of multiple heuristic strategies such as randomly selecting the initial solutions and applying random perturbation operations can help the algorithm avoid falling into local optimal solutions and increase its robustness. By introducing memory elements and combining multiple neighborhood structures, the search space and exploration ability of the algorithm can be improved. And using parallel multi-heuristic searches can accelerate the search speed and improve the convergence speed and accuracy. Erdogan et al. [131] selected 1-opt, 2-opt, and 2-exchange and other LS operators for the mixed-integer second-order conic (MISOC) model to improve the solution sequence. Ha et al. [132] developed ILS algorithms by introducing a splitting method, an adaptive perturbation operator, and an improved LS operator. The ILS algorithm developed by Luo et al. [133] includes a three-phase evaluation method, a special LS operator, and six perturbation operators. Wang et al. [134] used greedy heuristics to generate an initial solution and a 2-opt search operator to further optimize the solution. Saleu et al. [32] added DP to the framework of the ILS algorithm and used the DP to split the initial solution. The solution was iteratively improved and updated during the search process.

4.6. Variable neighborhood search

Variable neighborhood search (VNS) is an improved local search algorithm consisting of a variable neighborhood descent (VND) search operation and a perturbation process. It has been successfully applied to the TDRP and has a good optimization effect on combinatorial problems. The global convergence of the algorithm is reflected in its process for constructing the neighborhood structure set, the design of the local search strategy, and the stopping criterion. Improvement methods can

start with the construction of the initial solution, the neighborhood structure set, the design of the local search strategy, and the stopping criterion.

The quality of the constructed initial solution directly affects the performance of the algorithm. A good initial solution can ensure that the algorithm can obtain the global optimal solution or near-optimal solution in a short time. Two constructive methods are available, the random strategy and the heuristic strategy. The set construction principle is to ensure that the algorithm is as global as possible. The variable design of the local search process helps the algorithm to jump out of local optimal solutions. The selection of the stopping criterion has a direct impact on the global convergence and timeliness of the algorithm. Schermer et al. [135] used a parallel saving heuristic to generate initial solutions and added TS components to the VNS framework to prevent it from getting stuck in local optima. Kuo et al. [136] considered eight different neighborhood operators and efficiently explored the search space through a penalty function.

VND omits the random search process of VNS, replaces the random search and LS with a neighborhood structure search, and trades solution quality for a reduced solution time. It has the advantages of easy implementation and high computational efficiency. Studies to improve this approach have focused on neighborhood search methods. Ostermeier et al. [142] continuously improved the tours by optimally selecting a new set from the neighborhood of all existing tours based on the VND framework, which is called set improvement neighborhood search (SINS). De Freitas et al. [143] used an MIP solver to obtain the initial solution and then used the randomized VND (RVND) algorithm with seven neighborhood operators for LS. Gu et al. [144] embedded a hierarchical-based evaluation method and a segmentation-based ILS method into the VND framework, effectively exploring the neighborhood structure. Xu et al. [145] integrated SA into the RVND and formulated four specific neighborhood search strategies for the given problem. Campuzano et al. [146] employed a greedy algorithm to create an initial solution and designed three new neighborhood structures for iteration in the VND and multi-neighborhood TS algorithms.

General VNS (GVNS) is a widely adopted variant of VNS, incorporating VND during the local search phase. Ibroska et al. [140] applied sequential VND in the local search phase. Morim et al. [141] considered robot station assistance based on VRP with drones (VRPD) and proposed a GVNS that integrates five types of neighborhood structures to fulfill synchronization requirements.

4.7. Adaptive large neighborhood search

Adaptive large neighborhood search (ALNS) has good applicability in terms of solving TDRP. It searches multiple neighborhoods of the current solution to find a more satisfactory solution. With the expansion of the neighborhood search process, the algorithm does not easily fall into local optima, but its convergence speed decreases, and its time cost increases. Improvements can be made with respect to initial solution designs, destruction and repair operators, and local search strategies.

An improved strategy for generating the initial solution is first analyzed. The solution efficiency of ALNS is affected by the quality of the initial solution. Therefore, different strategies can be designed to generate effective initial solutions. Tu et al. [147] used a variety of heuristics to generate an initial solution and designed different destruction and repair operators to iteratively improve the solution quality. Sacramento et al. [148] used the NN algorithm to construct an initial solution, extended the continuation of SA during the iterative process, and defined two types of destruction operators and four types of repair operators.

Once initialization is complete, an adaptive layer can be introduced to allow the algorithm to perform intelligent operator selection, dynamically adjust operator weights, and use destruction and repair operators to process neighborhoods. By repeatedly designing LS strategies to explore new neighborhoods, the algorithm is prevented from

falling into local optima. Coindreau et al. [149] added removal heuristics and insertion heuristics to ALNS to explore neighborhood structures. Mara et al. [150] built a framework based on ALNS by using random initialization, iterative destruction, and repair operators to improve the obtained results. Li et al. [152] used a piecewise linear function and an adaptive probability mechanism to design a new ALNS. Wu et al. [159] embedded several different neighborhood search strategies into the improved ALNS framework. Jiang et al. [153] designed a collection of destroy and repair operators which were elaborately tailored. Meng et al. [158] embedded five destroy operations, five repair operations, and several preprocessing procedures into the ALNS framework. Zhao et al. [151] proposed a three-phase ALNS algorithm, which first uses removal and insertion operators to improve the initial solution, then finds the optimal robust schedule based on the grid search method, and finally determines the optimal solution through a LS. Nguyen and Ha [165] introduced deletion and insertion operators to the LNS framework to destroy and reconstruct the initial solution and designed an efficient movement evaluation procedure based on topological sorting to address synchronization constraints. Mulumba et al. [157] addressed the VRPD allowing for pickup and delivery, using diverse selection principles to destroy and repair different segments of the solution.

Other algorithms can also be integrated into the ALNS framework. Yan et al. [162] combined ALNS with the shortest path algorithm and used the scanning algorithm to construct an initial solution. Yan et al. [163] further improved the ALNS-SP algorithm and obtained the neighborhood of interest using multiple destruction and repair operators. Xia et al. [154] used a TPH based on a cost savings strategy to obtain an initial solution and then designed an ALNS algorithm to iteratively improve the result. Xia et al. [160] considered pickup and delivery, devising several effective destroy and repair operators for exploring the solution space, and introduced an SA criterion to prevent convergence to local optima. Xiao et al. [161] incorporated road grade into the operational standards and devised a novel partial-route removal operator along with two types of nearby insertion operations tailored to account for road steepness. Chen et al. [164] introduced tricycles into the TDRP and proposed a multi-start improved ALNS (MS-IALNS) by combining ALNS and ACO search mechanisms.

4.8. Greedy randomized adaptive search procedure

Greedy randomized adaptive search procedure (GRASP) is a multi-start algorithm for solving TDRP. It consists of two phases, construction and LS. The optimal replacement solution is found in the neighborhood of the feasible solutions in a continuous iterative manner. This approach can be improved in two aspects, the initial solution generation process and the neighborhood diversity, so that it achieves excellent convergence stability and strong global search capability.

Efficient heuristics can be designed to improve the quality of the initial solution. Marinelli et al. [167] constructed initial solutions based on a novel greedy heuristic. Ha et al. [166] used three different heuristics to generate initial routes and employed a splitting algorithm to improve the quality of the solutions.

The neighborhood structure can also be extended, including through the design of different neighborhood types and LS methods to better explore the search space. Almuhaideb et al. [168] designed two LS methods, hill-climbing LS and LS with SA, to adaptively select search neighborhoods. Vu et al. [169] used greedy randomized heuristics to generate initial solutions and designed several local search operations to improve the feasible solutions.

5. Approximation algorithms

Within TDRP research field, approximation algorithms (AA) are gaining traction as a means of efficiently deriving near-optimal solutions for combinatorial optimization challenges. These algorithms are particularly adept at providing approximate answers in polynomial

time, even when dealing with large problem inputs. This ability is crucial for solving complex logistical problems where exact solutions are computationally prohibitive. Related research on approximate algorithms is shown in Table 4.

The continuous approximation algorithm (CAA) stands out as a powerful tool for tackling TDRP, combining the precision of mathematical rigor with the practicality of polynomial solution times. This approach transforms the continuous optimization challenge into a series of quadratic programming problems, strategically iterating toward an optimal solution. Unlike exact algorithms, which can have exponential solution times for large problems, the CAA operates in polynomial time, making it more scalable and efficient for larger inputs. It also requires rigorous mathematical proofs to validate the quality of its solutions, distinguishing from heuristic algorithms that rely on experience-based approximations without requiring such rigorous verification.

The effectiveness of AA, including CAA, lies in its ability to balance accuracy with computational feasibility. By employing strategic search strategies and refining approximation functions, these algorithms improve both the accuracy of the solutions and the speed at which they are obtained. As such, continuous approximation is a sophisticated alternative to traditional exact and heuristic methods, offering higher search accuracy and solving efficiency for complex routing problems. This blend of qualities makes CAA a valuable asset in the optimization toolbox, especially for applications where time and solution quality are of the essence.

Tokekar et al. [170] presented an approximation algorithm with an approximation ratio of $\left(\frac{r_{max}}{r_{min}}\right)$, where r_{max} and r_{min} are the maximum and minimum radii of the sampling area, respectively. Nishira et al. [171] adopted a linear or quadratic function to approximate the objective function. Morandi et al. [172] proved that it is impossible to find a polynomial-time heuristic that can approximate the metric TSP with multi-drone (TSP-mD) within a constant factor, unless the $P = NP$. Wang et al. [173] proposed a polynomial-time approximation scheme (PTAS) to derive the general $\frac{1+\epsilon}{3+\epsilon}$ approximation ratio. Ulmer and Thomas [174] introduced a policy function approximation (PFA) method based on geographic districting, that approaches the MDP model by focusing on restricted policy classes and using a first-in first-out (FIFO) heuristic to facilitate real-time decision-making. Li et al. [175] utilized a derived CAA to approximate a multi-objective synthesis model as a simple function and used hexagons to divide the object region.

6. Artificial intelligence algorithms

In recent years, artificial intelligence (AI) algorithms have played an increasingly important role in many operational optimization problems. By gradually integrating AI into the TDRP, the efficiency and quality of solving complex problems has been significantly improved. Building on the foundations of computer science and statistics, AI employs sophisticated techniques such as deep learning (DL) and reinforcement learning (RL) to simulate human intelligence and decision-making processes. These methods offer significant advances in intelligent path planning and management capabilities, enabling more efficient and real-time solutions. Related research on AI is shown in Table 5.

DL, one of the primary machine learning (ML) methods used in AI, employs a variety of architectures, including but not limited to convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformer models, to process and learn from large datasets. Depending on the task, different architectures such as encoder-decoder frameworks or other network designs are applied to optimize learning and task performance. By optimizing the model parameters through backpropagation against an objective function, DL achieves optimal parameter combinations for tasks like classification and regression. However, despite its effectiveness, DL faces challenges such as long computation times and poor interpretability, which can be mitigated by

Table 4

Literature overview of approximate algorithms.

Alg	Reference	Appli	Pro	Mod	*T	*D	Obj	Truck visit	Drone visit number	Different L-R
AA	[170]	Oth	Sampling TSPN	-	1	1	CT	N	M	Y
AA	[171]	Log	FSTSP	IP	1	1	CT	Y	1	Y
AA	[172]	Log	TSP-mD	MILP	1	M	CT	Y	M	Y/N
PTAS	[173]	Log	PTP	IP	1	M	Makespan	N	1	Y
PFA	[174]	Log	SDDPHF	MDP	M	N	Delivery quantity	Y	1	N
CA	[175]	Log	-	CA	M	N	CD	N	1	N

Table 5

Literature overview of artificial intelligence algorithms.

Alg	Reference	Appli	Pro	Mod	*T	*D	Obj	Truck visit	Drone visit number	Different L-R
RL	[176]	Log	TDCDP	MIP	1	1	CT	N	M	N
RL	[177]	Log	TSP-D	MDP	1	1	CT	Y	1	Y
RL	[178]	Log	MT-VRPD	MILP	M	N	CT; CC	Y	1	Y
DRL	[179]	Log	FSTSP-STT	MDP	1	1	CT	Y	1	Y
DRL	[180]	Log	TSP-D	MIP	1	1	CC	Y	1/M	N
DRL	[181]	Log	T-DDP	MDP	1	M	CT	N	1	Y
DQL	[182]	Log	SDDVD	MDP	M	M	Delivery quantity	Y	1	Y
MARL	[183]	Log	T-DDP	M	N		CT/Profit	N	M	N
ML	[184]	Log	PLTSPHD	MIP	1	M	CC	N	1	N

increasing the dataset size, refining the neural network architecture, and integrating prior knowledge. Wu et al. [176] combined an encoder-decoder framework with RL by using a Monte Carlo policy gradient approach to train the model and adopting a parameter vector learning strategy.

RL, on the other hand, approaches the TDRP by modeling it as a Markov decision process (MDP), where an agent learns optimal policies through interactions with the environment and feedback in the form of rewards. Rather than relying on predefined mappings, RL focuses on exploration and exploitation to progressively refine its policy over time. Constraints such as time windows and traffic conditions are integrated into the state representation and reward function, guiding the agent's learning and decision-making processes. RL's ability to adapt strategies through multi-agent interactions and trial-and-error makes it particularly effective for tackling nonlinear and non-convex problems. The ongoing refinement of strategies in RL improves the practicality and robustness of path planning solutions. Kong et al. [178] introduced a self-driven RL approach for solving multi-objective optimization problems, integrating a graph neural network with an attention decoder to dynamically update node information.

Together, DL and RL represent two key AI methodologies that significantly enhance approaches to solving TDRP. While RL leverages environmental states, action spaces, and reward functions to iteratively learn optimal strategies, DL excels in extracting patterns from large datasets through neural network architectures. By combining these complementary strengths, DL and RL push the boundaries of what is possible in logistics and transportation management. Ongoing development and refinement of these algorithms promise further advances in solution efficiency, quality, and the overall intelligence of routing solutions.

Deep reinforcement learning (DRL) typically uses methods such as deep Q-networks (DQN) to approximate Q-values for state-action pairs, or other approaches to estimate state values (V-values). In addition to deep Q-learning (DQL), policy gradient methods—such as Actor-Critic (AC) and proximal policy optimization (PPO)—as well as hybrid methods like deep deterministic policy gradient (DDPG) are widely applied in DRL. Moreover, state space simplification in DRL is often achieved through techniques like feature selection, state abstraction, or state aggregation. By training on extensive interaction data with the environment, DRL models gradually learn to make intelligent decisions in complex environments. Chen et al. [182] proposed a DRL algorithm that uses information from both the state and action spaces. Arishi et al.

[184] developed a two-phase ML framework, in which the first phase adopts an unsupervised constrained k-means algorithm to reduce the complexity and size of the problem, while the second phase employs a DRL model that is trained on an unlabeled training set to generate feasible solutions in real-time. Bogrybayeva et al. [177] proposed a new end-to-end learning model based on the attention encoder and the long short-term memory (LSTM) network decoder. Wang et al. [180] presented a model consisting of the transformer-GNNs and an LSTM encoder. Liu et al. [179] adopted RL algorithms such as DQN and AC to solve the MDP model with uncertain factors. Bi et al. [181] compressed the action space and decision-making quantity to improve DQN learning efficiency. Bi et al. [183] implemented the MPE framework and multi-agent proximal policy optimization (MAPPO) to solve the multi-objective tuck-drone hybrid delivery problem and enhanced convergence by applying Monte Carlo Tree Search (MCTS) and reward normalization techniques.

7. Datasets

During the process of studying the TDRP, some representative test datasets have gradually been formed and accumulated, which can support verifications and analyses of the quality and effects of mathematical models and solving algorithms. Currently, there are two primary dataset generation methods, and representative datasets are provided as illustrations.

7.1. New generated dataset

The dataset in Murray and Chu [61]: A test dataset was constructed containing 72 instances based on the characteristics of the FSTSP. According to the rule that 80–90% of node data meets the access conditions of drones, each instance contains 10 nodes that have geographic data and demand data. The warehouse location is randomly selected as the mean of all coordinates, the mean of x-coordinates, or the origin. Then, the paper also considered the parallel drone scheduling TSP (PDSTSP), and 240 test instances were constructed in the same manner. Each instance contains 10 or 20 nodes, and 20%, 40%, 60%, or 80% of the nodes are located within the flight range of the drone.

The dataset in Agatz et al. [54]: Four sets of test instances were randomly generated with different geographic data distributions according to the characteristics of the TSP-D, ranging from 5 to 375 nodes. For the uniform instance, the node positions are uniformly distributed in

[1,100]. For the one-center instance, the normal distribution and deviation angle were used to simulate the distribution of circular cities. The two-center instance added the deviation distance and deviation probability based on the one-center instance. The restricted instances constrained the flight range and forbidden access nodes of drones.

The dataset in Ha et al. [166]: A test dataset containing 65 instances was randomly created in square areas of 100 km², 500 km², and 1000 km² according to the features of TSP-D. The dataset includes 5, 30, and 30 instances with 10, 50, and 100 nodes, respectively, and the number of clients generated per unit area is limited by density.

The dataset in Murray and Raj [84]: A test dataset comprising 100 instances was constructed based on the characteristics of the multiple FSTSP (mFSTSP), with 5, 8, 10, 25, and 50 nodes each having 20 instances. Among each group of 20 instances, 10 are generated from the map of Buffalo (a small city), and 10 are generated from the map of Seattle (a large city). For each 10 instances, five warehouses are located in the central location, and five warehouses are located in the outer locations.

The dataset in Kitjacharoenchai et al. [185]: A test dataset was devised consisting of 185 instances according to the characteristics of the multiple TSP-D (mTSP-D), including 150 medium instances with 25 and 50 nodes and 35 small instances with nine nodes. On a 1000*1000 grid, according to the distribution of nodes on the map, the dataset can be divided into five types of problems. The nodes of types 1–2 are evenly distributed in the plane, and the warehouses are located at (500,500), and (500,0). The warehouses of types 3–5 are all located at (500,500), but the nodes of type 3 are evenly distributed in a circular area with a radius of 500. The nodes of type 4 are evenly distributed in a circular area of size [300,500], and the nodes of type 5 are arranged in a cluster distribution within a circle with a radius of 125.

7.2. Improved or expanded datasets

The second type of dataset includes extensions of benchmark datasets, including datasets such as TSPLIB and Solomon. TSPLIB contains 14–85,900 nodes. Solomon created three types of instances with a random distribution, a cluster distribution, and clustering and random mixing. The advantage of the former is that it contains a large amount of node data, and the advantage of the latter is that its node data distribution is diversified. The method used to improve a benchmark dataset is as follows: some data in the benchmark dataset, such as geographic data, demand, or time window, are inherited. Then, the specific data types are modified according to the characteristics of the given problem. For example, when applied to different domains, different distribution ranges are used for the demand data of the visited nodes, because while node demand has little impact on the problem, the demand data can all be assumed to be equal.

The dataset in Freitas et al. [139]: According to the characteristics of the FSTSP, Freitas et al. [139] introduced the TSPLIB dataset and selected 25–51 nodes from it to form a test dataset containing 200 instances. The location and demand data were modified to ensure that 85–90% of the nodes could be served by drones.

The dataset in Luo et al. [128]: Three test instances were adopted from the Solomon benchmark dataset, namely, C101, R101, and RC101, according to the multi-visit TSP with multiple drones (mTSP-mD) characteristics. Then, the node service time in C101 was modified from 90 to 10, with the other two instances remaining the same. A test dataset containing 216 instances was generated. Six types of nodes are included, i.e., 8, 10, 15, 25, 50, and 100. The number of drones is limited to 1, 2, 3, or 4. And capacities of the drone are divided into three types: lower, medium, and high.

The dataset in Tu et al. [147]: 60 instances were selected with 50 and 100 nodes from the work of Ha et al. [166] according to the TSP-mD characteristics, and all parameters were inherited from the benchmark dataset.

The dataset in Cavani et al. [186]: A subset of data was chosen from

the Poikonen et al. [20] benchmark dataset according to the characteristics of the TSP-mD and formed a test dataset containing 40 instances with 10, 15, 20, and 25 nodes. The nodes that must be filled are randomly generated on a 50*50 grid, and the coordinates of all nodes are evenly distributed in the two-dimensional grid.

The current commonly characteristic datasets are shown in Table 6.

8. Applications

Cooperation between trucks and drones enables rapid response and efficient execution of tasks, and has been widely applied in various fields, playing a significant role in improving work efficiency. Based on the application scenarios mentioned in existing literature, we have conducted a review of several typical domains. However, it should be noted that due to the differences between professions, certain industry-specific terms may have alternative expressions, despite sharing similar meanings in essence.

8.1. Logistics

In the field of logistics (also known as delivery), the collaboration of trucks and drones has emerged as an important solution in last-mile delivery, driven by the rapid development of drone technology and e-commerce. With advances in addressing the last-mile delivery problem, the scope of items to be delivered has expanded from conventional goods such as retail products and postal packages to include specialized items such as food and pharmaceuticals.

Drones are capable of delivering quickly across various terrains, especially in remote areas and emergencies, thereby enhancing delivery efficiency. Meanwhile, trucks have the advantages of transporting more goods, covering a wider area, and meeting large-scale, long-distance logistics needs. Additionally, trucks can serve as charging centers, powering-up drones and delivering goods to certain demand points. In urban areas such as commercial districts and residential neighborhoods, trucks often encounter severe traffic congestion, while the demand for timely delivery satisfaction from customers continues to rise. Drones offer a superior solution to meet these time-sensitive requirements. In terms of convenience, drones surpass traditional trucks. Through the establishment of appropriate delivery interfaces, customers may not even need to be physically present during the delivery process. In remote regions with inadequate transportation infrastructure, trucks may fail to meet the demands of logistical operations. In such scenarios, drones can efficiently handle item transportation. During natural disasters or emergency situations, ground transportation systems might be impeded, thereby highlighting the importance of drones in conducting aerial material rescue operations.

Table 6
Overview of typical datasets.

Mode	Reference	Problem	Number	Composition	Types
One-truck and One-drone	[61]	FSTSP	72	Generation	Demand: 10
	[54]	TSP-D	30	Generation	1-center; 2-center
	[166]	TSP-D	65	Generation	10, 50, 100
	[20]	TSP-D	200	Generation	10, 15, 20, 48
One-truck And Multi-drone	[139]	FSTSP	200	Improvement	TSPLIB
	[147]	TSP-mD	65	Improvement	[166]
	[186]	TSP-mD	40	Improvement	[20]
	[84]	mFSTSP	100	Generation	5, 8, 10, 25, 50
Multi-truck and Multi-drone	[187]	VRPD	102	Improvement	[61]; [54]
	[185]	mTSPD	185	Generation	9 (35); 25\50 (150)
	[128]	mTSP-mD	216	Improvement	Solomon
	[61]	PDSTSP	720	Generation	10, 20
	[188]	PDSTSP	240	Improvement	TSPLIB

Meituan has made significant progress in efficient and convenient food delivery through its drone delivery services. Starting from August 2, 2023, the Changbai Mountain Resort in China has offered drone meal delivery services, and Meituan has expanded its services to cover not only scenic areas but also remote camping sites [189]. Drones are capable of autonomously planning flight paths and delivering orders accurately to the nearest designated landing platforms. Furthermore, Meituan strategically installed launch pads on the rooftops of shopping centers, near residential areas, and office buildings to facilitate drone takeoff. Over the past year and a half since Meituan launched its first drone delivery route in Shenzhen, they have successfully fulfilled over 184,000 orders and established 17 delivery routes nationwide [6]. This point-to-point delivery approach ensures that drones can navigate along predetermined routes and safely deliver food to customers.

Research on TDRP for medical supply delivery is limited, focusing mainly on emergency aid, and small-scale deliveries with low environmental sensitivity. The unique nature of medical supplies and strict storage and transportation requirements pose higher technical and managerial challenges for broader applications [190]. Particularly, the coordination of trucks and drones for complex medical logistics remains underexplored. Medical applications require highly reliable, secure, and timely delivery systems, requiring customized solutions that current models do not fully address. The paper highlights the significant potential of TDRP in enhancing emergency medical deliveries and recommends future research to develop models specific to medical logistics. It also emphasizes the need for interdisciplinary collaboration to ensure that proposed solutions meet the needs of the healthcare sector.

In recent years, the evolution of delivery problems has led to the emergence of numerous additional challenges. For example, Karak et al. [69] and Jiang et al. [153] considered pickup and delivery problems simultaneously, Mara et al. [105] included charging stations for electric vehicles along the routes, and Teimoury and Rashid [23] assumed that customers exist with probability. Scholars like Lin et al. [191] and Zhang and Li [192] have conducted relevant research on the distribution efficiency of the contactless distribution of essential or special commodities during the COVID-19 epidemic.

Research on TDRP holds significant potential for revolutionizing real-world logistics and delivery systems and driving both technological innovation and industry transformation. In the near future, integrating trucks and drones is expected to greatly improve logistics efficiency and increase the flexibility of delivery networks. In particular, TDRP could transform the last-mile delivery, which currently lags behind other logistics processes like warehousing and sorting in automation and intelligence. By automating and optimizing the final stage of delivery, TDRP could substantially improve delivery speed, cost efficiency, and service quality. As technology advances, TDRP is expected to drive significant innovation in logistics systems, paving the way for more automated and intelligent delivery networks and ultimately transforming the logistics industry. However, despite the rapid growth of research in this area, practical implementation remains a significant challenge. Real-world circumstances, such as the need for recipient presence during drone deliveries to apartment buildings and the development of protocols for recipient absence, pose hurdles. Additionally, regulatory restrictions, such as the incomplete opening of low-altitude airspace in China and the ongoing development of drone policy, present pressing practical issues that need to be addressed.

8.2. Inspection in the civil

In the field of inspection, the integration of trucks and drones can be applied to various operations, including inspections of power lines, oil pipelines, buildings, and municipal roads. Drones are particularly useful in tasks that require large coverage areas, complex and dangerous site environments, or high-altitude operations, as they can achieve fast, efficient, and accurate inspections. For ground inspections and detailed checks, trucks can be used for ground monitoring [122]. The

cooperation of the two can greatly improve inspection efficiency and reduce labor costs. For example, in power line inspection, drones take aerial photos of transmission lines and substations, and transmit the data to the ground inspection truck, where the data is analyzed and processed.

As early as 2015, the State Grid Corporation of China adopted drones to inspect transmission towers in Jiangsu Province, completing the inspection within 10 minutes. Compared with 30 minutes for manual inspection, drones greatly improve inspection efficiency and reduce safety risks [193]. Currently, the use of drones for inspecting power towers and transmission lines is becoming a mainstream approach, replacing manual inspection. At present, all inspection tasks are performed in the air, and trucks are mainly used to transport drones and change the batteries. Based on the characteristics of the objects being inspected, inspection tasks are typically classified into point tasks, line tasks, area tasks, and mixed tasks. Huang et al. [95], Baik et al. [194] and Wang et al. [97] abstracted the inspection objection, the power tower, into a single task point. Line tasks typically cover larger areas. Amorosi et al. [195] modeled line tasks as graph. When drones perform area tasks, it is necessary to plan their external flight paths as well as set the scanning patterns for internal access. Mixed tasks are composed of the above types of tasks. Liu et al. [67] established a two-layer point-arc routing problem with point tasks and line tasks.

8.3. Patrol in the civil

In the field of patrolling (also known as monitoring), research on truck-drone coordination is primarily applied to security, border, and traffic patrols in large-scale complex environments, aiming to improve patrol efficiency and safety. By real-time communication such as the internet and LAN, drones and trucks can share data and cooperate with each other. Through coordinated cooperation in the air and on the ground, comprehensive, timely, and high-precision patrol operations can be achieved.

In May 2016, the city of Zhanjiang, Guangdong Province, China, pioneered the use of drones to patrol key waterways and sea areas. The drone was mainly deployed for tasks such as monitoring quays, water channels, offshore construction, and ship sewage disposal. It was capable of capturing 4K high-definition panoramic photos and transmitting real-time data to the base. Additionally, the relevant authorities have planned to expand the application areas of drones in the future, including shallow-water construction, offshore access order-keeping, and ship anchoring orders [196]. At the same time, traffic police in Guangzhou have also used drones to conduct aerial patrols on highways [197].

Unlike power inspection, patrol tasks occur on the road network. Both drones and trucks can participate in these tasks to complete them. For instance, during traffic patrols [145], drones can monitor real-time traffic conditions and capture images, while trucks supervise and manage traffic flow, accidents, and illegal behaviors on the road. In forest patrols [59], drones fly to areas inaccessible by trucks for monitoring, meanwhile, trucks patrol on forest roads, monitor the status of natural resources, and provide early warnings regarding forest safety.

8.4. Intelligence, surveillance, and reconnaissance in the military

In the military field, combined research on trucks and drones is mainly used for intelligence, surveillance, and reconnaissance (ISR). In terms of intelligence collection, drones provide a large-scale monitoring perspective through high-altitude photography, while trucks can search for targets on the ground and provide more accurate positioning information [29]. With respect to combat, drones provide long-range attacks against enemy targets through aerial attacks, while trucks can launch attacks and provide more intense fire support. As regards military supply, drones can quickly transport materials and supplies through aerial flight, while trucks can perform transfer and transportation tasks on the

ground. Regarding surveillance, drones can obtain large-scale data by flying at high altitudes and covering areas with obstacles, while trucks provide detailed surveillance of specific areas on the ground [104,163].

8.5. Others

The collaboration of trucks and drones also has potential applications in other fields. This combination can realize monitoring, law enforcement, and emergency response, thereby enhancing the intelligence and automation of environmental protection work [104]. At the sites of disasters such as fires, earthquakes, and floods, the use of trucks and drones can assist with monitoring, surveying, and rescue efforts, resulting in improved rescue efficiency and safety [198]. In crop production, they can enable intelligent diagnosis and precise pesticide application [170], leading to enhanced precision and efficiency of agricultural operations, as well as reduced pesticide waste and crop losses.

In summary, TDRP has broad application prospects in various fields, which provides strong support for the research and development and practical application of modern technology. In addition, the practical demands and achievements of real-world applications have provided valuable insights and directions for innovative research on the problem. As technology continuously advances and application scenarios expand, the range of potential applications for truck-drone combinations will broaden and deepen, bringing more opportunities and challenges to all aspects of various industries.

9. Discussion

This section will provide a systematic discussion of the overall content, including approaches, applications, the gaps between theory and practice, as well as a multidisciplinary perspective.

9.1. Approaches: characteristics, applicability, innovations, limitations, and improvements

In this section, we explore the characteristics, applicability, and limitations of various approaches used in solving TDRP. While each approach presents innovative features, it also faces inherent limitations, which have driven the need for theoretical innovation to overcome specific challenges. It's important to note that these algorithms do not inherently differ in quality, but their effectiveness varies based on the context of the problem being addressed. Our analysis will critically evaluate their impact on TDRP research and practical applications.

Exact algorithms are highly effective for small-scale, well-structured problems due to their ability to guarantee optimal solutions with precision, making them ideal for tasks such as logistics and inspection involving one-truck and one-drone or one-truck and multi-drone configurations. While traditionally limited to smaller problems, recent innovations in innovative decomposition methods and integrated hybrid models have extended their applicability to larger, more complex scenarios. However, their primary limitation lies in the exponential growth of computational cost with increasing problem size, often making them impractical for large-scale applications. To address this, hybrid models combining exact algorithms with heuristic or metaheuristic algorithms, along with problem decomposition or pruning strategies, can reduce the computational space and enhance scalability, making exact algorithms more feasible for complex problem-solving.

Heuristic algorithms efficiently construct feasible solutions by exploring the set of a given problem in a limited number of steps, making them well-suited for complex or large-scale problems that require rapid response, such as one-truck and multi-drone or multi-truck and multi-drone configurations. Their innovation lies in dynamic adjustment mechanisms, local search strategies, and multi-level searches, which enhance their ability to quickly generate near-optimal solutions. But they often get trapped in local optima, compromising solution quality.

This limitation can be mitigated by introducing diverse initial solution generation strategies or integrating global search techniques to expand the exploration of the solution space and improve overall solution quality.

Metaheuristic algorithms, combining stochastic processes and LS methods, are highly flexible and adept at solving complex routing problems, such as multi-truck and multi-drone configurations. And the previous analysis indicates that metaheuristic algorithms have garnered the most attention. Their use of random strategies enables extensive exploration of the solution space, enhancing the likelihood of finding near-global optimal solutions. With innovations like multi-objective optimization and adaptive mechanisms, they excel in complex environments, balancing resources and time effectively. Their high computational requirements, especially in multi-objective tasks, pose a challenge. Enhancing algorithm parameters and incorporating efficient adaptive mechanisms can improve their performance and reduce computational intensity.

Approximation algorithms provide scalable, efficient solutions for large-scale TDRP, and their polynomial-time performance makes them ideal for resource-constrained environments. Their use of sensitivity analysis helps to optimize model parameters and improve adaptability. While they excel in computational efficiency, their solution quality may not meet the precision required in critical scenarios and require strict mathematical proofs for quality. Improving their structure could enhance accuracy, ensuring a better balance between efficiency and precision for more reliable results in demanding applications.

AI algorithms demonstrate strong adaptability in real-time decision-making in dynamic and uncertain environments, learning and predicting future demands and environmental changes. They are particularly effective for real-time decision-making in complex TDRP scenarios, such as multi-truck and multi-drone routing problem. While they may not always match the precision of metaheuristic algorithms, AI methods significantly reduce solution time in large-scale problems due to RL and parallel computing. Their reliance on large training datasets and sensitivity to data quality, along with complex and time-consuming training processes, pose challenges. Improving learning efficiency, optimizing model structures, enhancing data preprocessing, and incorporating domain knowledge can increase their robustness and adaptability in addressing complex routing challenges.

9.2. Applications: importance, challenges, and opportunities

The literature reviewed in this paper shows that TDRP applications are primarily focused on logistics, particularly parcel and last-mile delivery, driven by the industry's pressing need to optimize delivery times and reduce costs. Logistics provides the ideal environment for TDRP because of the high demand for route optimization and the ability to deploy trucks and drones in complementary roles. The well-established infrastructure and growing use of autonomous systems in logistics make it the most mature and widely explored domain for TDRP. In other fields, such as power inspection and agricultural monitoring, TDRP has distinct advantages like precision in remote area coverage and the ability to handle repetitive, large-scale tasks. However, these scenarios have attracted less research due to the higher application complexity and detail of these environments. The expanding diversity of TDRP applications introduces new challenges for algorithm design, requiring solutions that account for varying complexities and real-time adjustments. The maturity of the application scenario directly influences the choice of algorithms, with logistics and ISR necessitating more dynamic, adaptive approaches.

Despite its promise, TDRP faces several practical challenges. In logistics, ensuring safe drone flight in dense urban areas and managing peak demand periods with large order volumes are significant obstacles. The complexity of urban environments, including air traffic and infrastructure, necessitates robust and adaptive algorithms capable of real-time decision-making. Fields like power inspection and agricultural

monitoring are limited by the simplicity and predictability of the environment, leading to less research focus in these areas. In ISR operations, TDRP is challenged by rapidly changing battlefield conditions, adversarial interference, and the need for real-time adaptability. Many studies focus on specific, controlled scenarios, and addressing uncertainty in dynamic environments remains a key research gap. Future work should derive research questions from real-world industry challenges, using generalized methods to broaden the scope of TDRP applications and improve solution robustness.

To improve the application of TDRP across diverse fields, the focus should be on developing efficient and robust algorithms that can adapt to dynamic and unpredictable environments. This involves accurately translating models, designing effective heuristic rules, and integrating AI frameworks that enhance solution adaptability. Rigorous testing and evaluation of algorithms in real-world scenarios are essential to verify their effectiveness. Additionally, hybrid models that combine heuristic, metaheuristic, and AI approaches can help balance efficiency and precision. By ensuring that algorithms are both flexible and capable of handling uncertainty, TDRP solutions can become more practical and widely applicable across different sectors such as logistics, power inspection, and agricultural monitoring.

9.3. Gaps between theory and practice

Despite the rich approaches provided by theoretical research for TDRP, there are significant gaps exist between theory and practice due to several factors.

One key gap results from the contradiction between the complexity of real-world problems and the simplified nature of theoretical research. Theoretical studies often focus on abstract, simplified models with fixed processes, stable truck and drone capacities, and a limited scope of objectives, such as optimizing completion time or reducing time costs. In contrast, practical implementation requires consideration of broader economic factors like equipment costs, maintenance, and operator training. This gap between theoretical models and real-world conditions can limit the practical applicability of research results, as the more complex and dynamic nature of real-world scenarios requires adaptable and comprehensive solutions.

Another significant gap is that the static or predictable parameters used in theoretical models do not capture the uncertainty of real-world conditions. Real-world applications face fluctuating demand, weather changes, traffic conditions, and unforeseen obstacles, all of which introduce uncertainty into TDRP. Theoretical models often assume that key information is either known or can be predicted with certainty, but in practice, these dynamic factors create a need for algorithms that can adapt to changing conditions in real time. This gap highlights the limitations of current theoretical models when applied to the unpredictable nature of real-world environments.

A third gap is the lack of integration of regulatory and policy considerations into theoretical research. In practice, the use of drones is subject to strict regulations regarding flight altitude, operational zones, safety, and privacy, which significantly impact the feasibility of TDRP. These regulations are critical in real-world applications, but theoretical studies often overlook them or fail to incorporate them into the models. This leads to solutions that may be theoretically optimal but impractical or illegal under current regulations. To close this gap, future research must account for the legal and regulatory frameworks that govern drone operations.

Practical applications drive the refinement of theoretical research, while theoretical advances provide innovative solutions to real-world challenges. To bridge the gap between theory and practice, future research should prioritize addressing the complexities of real-world environments, dynamic uncertainties, and regulatory constraints. By focusing on these practical challenges, researchers can develop more robust and adaptable models that effectively translate theoretical insights into real-world solutions for TDRP.

9.4. Multidisciplinary research for comprehensive TDRP

Considering that TDRP is inherently interdisciplinary, examining existing research through the lens of multiple disciplines highlights both the complexity of the problem and the potential for innovation through cross-disciplinary approaches. Although TDRP originates from optimization problems, methodological innovations should not be confined solely to optimization. Instead, optimization should serve as the central thread, complemented and enhanced by other multidisciplinary approaches. TDRP not only involves operations research but also draws from computer science, control engineering, electrical engineering, robotics, communications, and even policy studies. By integrating these fields, researchers can develop more comprehensive models that better capture the real-world intricacies of TDRP, leading to more robust solutions. Each discipline contributes critical support to various aspects of TDRP, such as algorithm design, theoretical optimization, real-time control, energy management, and automated operations, thereby significantly advancing the development of the field.

Specifically, AI and ML from computer science optimize scheduling and decision-making, while operations research enhances resource allocation and path planning. Control engineering ensures flight precision and safety, and electrical engineering improves energy efficiency for longer missions. Robotics enables autonomous navigation, supported by communications for seamless data exchange between drones and trucks. Policy studies ensure regulatory compliance, making these interdisciplinary efforts essential for advancing the effectiveness and practical deployment of TDRP. The integration of these disciplines in TDRP research fosters the development of innovative, practical solutions that drive the field forward and enhance its effectiveness in real-world applications.

While individual methods can address localized problems, applying TDRP within a broader system requires a multidisciplinary approach to ensure its successful implementation. A relevant example is the study by Qiu et al. [199], which considered the problem of unreliable communication between drone systems, often subject to delay and loss. Wang et al. [200] investigated collision-free path planning for multiple drones in uncertain environments, employing multi-agent RL based on probabilistic dynamics to design routing strategies. Similarly, Yu et al. [201] introduced a distributed adaptive fuzzy formation control for multiple drones, incorporating the dynamic behavior of the drones to enhance system adaptability and control accuracy. Heidari et al. [202] further explored the advantages of ML in tackling challenges such as delay, throughput, interconnectivity, and reliability for drones. However, they also highlighted challenges like wireless medium unpredictability, high mobility, and limited battery life, leading to rapid topological changes. It can be observed that TDRP urgently requires the integration of interdisciplinary knowledge to provide a more comprehensive understanding of the practical challenges faced during implementation.

By integrating insights from multiple disciplines, future research can more effectively address real-world challenges, expanding the potential of TDRP solutions and driving innovation in this field. Multidisciplinary collaboration will be key to developing robust, adaptable solutions that can meet the growing complexity and demands of practical applications.

10. Future research directions

This section outlines future research directions for TDRP, drawing on the findings presented above. The focus lies on innovating problems and approaches, exploring large-scale application scenarios, bridging the gap between theory and practice, and integrating multidisciplinary techniques. These lines of research promise to drive significant progress, enhancing the practical implementation of TDRP in the real world.

10.1. Designing and innovating problems and approaches

The future of TDRP research depends on the innovation and design of

both problems and approaches, particularly in areas such as drone performance, multi-objective optimization, multi-mode integration, and the adaptation of algorithms with specific problem characteristics.

First, drone performance plays a crucial role in path planning. Factors such as endurance, battery capacity, speed, and payload significantly affect problem outcomes. However, existing studies often simplify these parameters, overlooking the effects of external conditions like wind and temperature. Future research could incorporate more realistic drone characteristics and refine energy optimization models to better reflect operational scenarios. As drone technology evolves, the capacity for multi-tasking and flexible take-off and landing will be gradually improved, and the existing operational constraints can be relaxed according to the development of relevant technology and infrastructure.

Second, the combination of multi-objective, multi-mode, multi-stage, and multi-factor optimization is the embodiment of the complexity of TDRP. While current research mainly focuses on single-objective optimization (e.g., minimizing time or cost), real-world applications require balancing multiple objectives. Future research should address these complexities through diversified multi-objective and multi-stage optimization, expanding combination modes, and integrating multiple factors, such as time windows, simultaneous pick-up and delivery, and multi-depot scenarios.

In addition, matching approaches with problem requirements is key to advancing TDRP research. For single-objective problems, hybrid optimization models combining exact and heuristic/metaheuristic algorithms are recommended to strike a balance between accuracy and efficiency. In multi-objective scenarios, algorithms based on Pareto optimality are crucial for simultaneously optimizing cost, time, emissions, and service quality. For dynamic environments, real-time optimization algorithms, particularly those using AI for adaptive path and task allocation, can handle variability. Research into parallel and distributed computing, especially for large-scale, multi-truck, multi-drone routing, could further enhance efficiency and scalability.

Lastly, optimizing for dynamic environments is essential. Real-world scenarios expose drones and trucks to fluctuating traffic, weather, and task demands. Future research should focus on developing dynamic path planning models that can adjust in real time based on environmental changes. These models, driven by real-time data, will enable smarter task execution.

By exploring these aspects, TDRP research can better address complex real-world applications, improving task scheduling and path optimization, while fostering innovation in algorithm design.

10.2. Exploring application scenarios

Future research should prioritize optimizing large-scale scenarios and implementing demonstration applications to accelerate the translation of theoretical results into practical applications.

Exploring large-scale TDRP scenarios is a vital step toward bridging the gap between theory and application. As the number of task points and the scale of drones and trucks increase, traditional small-scale path planning approaches become inadequate. Future studies should exploit distributed computing and parallel algorithms to partition path planning tasks across multiple computational nodes, improving efficiency in large-scale scenarios. Hierarchical optimization can also be employed to divide the whole into sub-regions for local optimization, with global coordination to reduce complexity and enhance scheduling efficiency.

The design and implementation of demonstration applications will provide valuable support for TDRP technology adoption. In areas like logistics, inspections, patrols, and ISR, TDRP can be applied to validate its effectiveness in real-world environments and provide critical data for further algorithm optimization. For example, coordinating drones and trucks in urban delivery can enhance efficiency in parcel and food distribution, while in inspections, patrols, and ISR, efficient path planning can ensure extensive area coverage. And in emergency response, TDRP's

responsiveness and planning efficiency in high-pressure environments will be tested.

Data-driven optimization is another critical direction for future research. As TDRP is deployed across various fields, it will generate vast amounts of operational data, which can be used to develop data-driven learning algorithms for continuous optimization. Through ML and real-time feedback, the coordination can adapt to complex environments, improving their robustness and flexibility.

At last, cross-industry collaboration and standardization are crucial for TDRP's widespread adoption. Future research should promote cross-sector collaboration to establish unified technical standards and shared platforms, integrating resources from different domains. By developing standardized technical frameworks, TDRP can ensure adaptability and security across industries, laying the groundwork for global deployment.

Through the optimization of large-scale scenarios, demonstration applications, data-driven optimization, and cross-industry collaboration, future research will bring TDRP closer to practical implementation, providing powerful technological support for its application in various fields.

10.3. Bridging the gaps between theory and practice

The gaps between theory and practice present a major challenge for TDRP applications. Future research needs to address several key areas to bridge this gap.

First, the complexity of real-world problems often exceeds the simplified assumptions of theoretical models. In practice, tasks are affected by dynamic variables such as traffic congestion, weather changes, and task prioritization. Future research should develop dynamic path optimization models capable of adapting to these changes, while integrating multi-objective optimization techniques to address the multiple demands of real-world scenarios.

Second, uncertainties and fluctuations, which are often difficult for theoretical models to capture, should be considered. Future research should incorporate stochastic and robust optimization techniques to handle variable conditions, such as drone endurance, battery levels, and weather. By leveraging IoT and real-time data streams, dynamic path planning can be made more responsive to uncertainty.

What's more, regulatory and policy constraints are frequently overlooked in theoretical research. Future studies should develop path planning algorithms that account for policy regulations, ensuring TDRP compliance across different jurisdictions. Cross-border logistics scenarios, in particular, require the integration of multi-national legal frameworks to facilitate compliant operations.

By addressing these areas, future research can significantly narrow the gap between theory and practice, driving the broader adoption of TDRP in both academic and commercial contexts.

10.4. Combining techniques and methods from multiple disciplines

The integration of multidisciplinary techniques offers new perspectives and opportunities for future TDRP research.

First of all, the diversification and standardization of datasets is crucial. TDRP spans various fields, yet existing datasets are often derived from a single application scenario, limiting their applicability. Future research should focus on diversifying and standardizing datasets by integrating data from multiple domains to build more generalizable, cross-disciplinary datasets. Standardized test datasets and benchmarks can also facilitate fair comparisons across algorithms, leading to smarter path optimization.

Then, building multi-layered simulation environments will enhance the realism of theoretical research. By combining drone flight data and truck traffic information, multi-layered simulation systems can reflect real-world dynamics more accurately. These environments, supported by physical simulation and digital twin technologies, can reflect the status and environment dynamics, and the task execution is optimized.

At the same time, virtual reality technology can enhance the visualization of system operation.

Interdisciplinary algorithm development will drive innovation in TDRP. Combining AI, ML, and robotics, future research can develop more intelligent path planning methods, increasing solutions' adaptability and efficiency. The combination of robotics and behavioral science will lead to more efficient solutions, especially when dealing with complex tasks and the collaboration of multiple unmanned systems. Behavioral science and economics can also be integrated to explore the economic benefits and user behavior associated with TDRP, further refining optimization strategies.

Furthermore, environmental sustainability and energy management are important considerations that cannot be ignored in future research. As environmental concerns grow, TDRP must prioritize not only efficiency but also energy consumption and environmental impact. By introducing factors like carbon emissions and energy optimization into objectives and constraints, research can develop greener, more sustainable solutions, increasing the overall environmental benefits of TDRP.

By focusing on intelligent, interdisciplinary solutions and adapting to complex real-world scenarios, future TDRP research will achieve greater problem-solving capabilities and broader application potential.

11. Conclusions

In conclusion, TDRP has received considerable attention since 2015, with notable achievements in both theoretical research and practical applications. The literature reviewed in this paper shows a growing scholarly interest in TDRP. This paper summarizes various algorithms, discusses several application scenarios, and suggests potential research directions based on the discussions, highlighting common challenges faced by these algorithms. Specifically, computational complexity and time efficiency limit the scalability of exact algorithms; in complex environments, the efficiency and solution quality of heuristic and meta-heuristic algorithms need to be more stable; dynamic environments and real-world constraints require greater robustness in algorithms; model construction should prioritize the practicality of solutions; and there is a need for standardized evaluation criteria and practical validation of algorithms. Additionally, better trade-offs in multi-objective optimization are needed. These challenges underscore the necessity for further research and innovation to develop more scalable, adaptable, and robust TDRP algorithms for practical implementation.

This paper provides essential guidance for both researchers and practitioners. For researchers, the review clarifies the current state of research, unresolved problems, and future research directions, recommending the selection of appropriate approaches for specific fields while emphasizing the impact of theoretical assumptions on practical applications. Interdisciplinary collaboration is highlighted as crucial for enhancing algorithm performance, particularly in the integration of

optimization, operations research, electrical engineering, and communication technologies. The review underscores the importance of the synergy between theory and practice, where challenges in real-world applications can inspire new research directions. For practitioners, the review identifies the potential and limitations of technological applications, advising the selection and refinement of algorithms based on specific industry needs, particularly in logistics, patrol, and ISR fields. By advancing both theory and practice in a cyclical manner, researchers and practitioners can jointly drive the progress and practical application of TDRP technologies.

This review highlights that research in the area is advancing into a new phase. As practical applications progress, increasingly complex problems are being identified and addressed, allowing researchers to gain considerable experience in model construction, algorithm design, and practical implementation. The results of this research have already been applied in certain real-world scenarios, demonstrating significant potential and value. However, despite this progress, the review emphasizes that current research still faces many challenges and limitations. For instance, the timeliness and robustness of existing algorithms in dealing with large-scale problems need further improvement, and the complexity and uncertainty of real-world environments have not been fully addressed. What's more, there remains a gap between research and practice, with the effectiveness and feasibility of theoretical models and algorithms in real-world applications requiring further validation and refinement. Therefore, future research should focus more on these practical issues to drive the development and application of TDRP.

CRediT authorship contribution statement

Jie Duan: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **He Luo:** Writing – review & editing, Writing – original draft, Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Guoqiang Wang:** Supervision, Resources, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported in part by the National Natural Science Foundation of China under grant 71971075, grant 72271076, and grant 71871079.

Appendix. List of abbreviations

Category	Abbreviation	Detail
Application	Ins	Inspection in civil
	ISR	Inspection, surveillance, and reconnaissance in military
	Log	Logistics
	Oth	Others
	Pat	Patrol in civil
Approach	AA	Approximation algorithm
	AC	Actor-Critic
	ACO	Ant colony optimization
	ACS	Ant colony system
	AI	Artificial intelligence
	ALNS	Adaptive large neighborhood search
	B&B	Branch and bound

(continued on next page)

(continued)

Category	Abbreviation	Detail
Model	B&C	Branch and cut
	BD	Benders decomposition
	B&P	Branch and price
	B&P&C	Branch and price and cut
	CAA	Continuous approximation algorithm
	CCG	Column-and-constraint generation
	CFRS	Cluster first, route second
	CG	Column generation
	CH	Constructive heuristic
	CNNs	Convolutional neural networks
	DDPG	Deep deterministic policy gradient
	DL	Deep learning
	DP	Dynamic programming
	DQL	Deep Q-learning
	DQN	Deep Q-network
	DRL	Deep reinforcement learning
	FIFO	First-in first-out
	GA	Genetic algorithm
	GRASP	Greedy randomized adaptive search procedure
	GVNS	General variable neighborhood search
	HGA	Hybrid genetic algorithm
	HH	Hybrid heuristic
	ILS	Iterated local search
	LS	Local search
	LSTM	Long short-term memory
	MA	Memetic algorithm
	MAPPO	Multi-agent proximal policy optimization
	MCTS	Monte Carlo Tree Search
	ML	Machine learning
	MMAS	Max-min ant system
	MPH	Multi-phase heuristic
	MS-IALNS	Multi-start improved adaptive large neighborhood search
	NN	Nearest neighbor
	NSGA-II	Non-dominated sorting genetic Algorithms-II
	O-ELGA	Odd-even layered genetic algorithm
	PFA	Policy function approximation
	PPO	Proximal policy optimization
	PSO	Particle swarm optimization
	PTAS	Polynomial time approximation scheme
	RFSS	Route first, split second
	RL	Reinforcement learning
	RNNs	Recurrent neural networks
	RTSP	Route-transform-shortest path
	RVND	Randomized variable neighborhood descent
	SA	Simulated annealing
	SINS	Set improvement neighborhood search
	SPH	Single-phase heuristic
	SPP	Shortest path problem
	TDCDS	Truck-drone cooperative delivery system
	TPH	Two-phase heuristic
	TS	Tabu search
	VNS	Variable neighborhood search
	VND	Variable neighborhood descent
	IP	Integer programming
	LP	Linear programming
	MDP	Markov decision process
	MIBP	Mixed integer bilinear programming
	MIP	Mixed integer programming
	MILP	Mixed integer linear programming
	MINLP	Mixed integer nonlinear programming
	MISOC	Mixed-integer second-order conic
Objective	CC	Completion cost
	CD	Completion distance
	CE	Carbon emission
	CS	Customer satisfaction
	CT	Completion time
	EC	Energy consumption
	TN	Truck number
	WT	Waiting time
Problem	CT	Conventional truck delivery
	CVTSP	Carrier-vehicle traveling salesman problem
	DAPDP	Drone-assisted pickup and delivery problem
	DO	Drone operations
	DT	Drone delivery supported by truck
	DTCO	Drone-truck combined operations
	EVRPD	Electric vehicle routing problem with drones

(continued on next page)

(continued)

Category	Abbreviation	Detail
Term	FSTSP	Flying sidekick traveling salesman problem
	GV-UAV	Ground-vehicle and unmanned-aerial-vehicle routing problem
	HTDDS	Hybrid truck-drone delivery systems
	MD	One-to-many delivery by drone
	mDPD-T	Multi-visit drone-assisted pickup and delivery problem with time windows
	mFSTSP	Multiple flying sidekick traveling salesman problem
	mTSP-D	Multiple traveling salesman problem with drones
	mTSP-mD	Multi-visit traveling salesman problem with multiple drones
	OD	One-to-one delivery by drone
	PDSTSP	Parallel drone scheduling traveling salesman problem
	RP-D	Drone-aided routing problem
	SARP-D	Share-a-ride problem with drones
	TDRP	Truck-drone routing problem
	TDPDP	Turck-and-drone pickup and delivery problem
	TmDTL	Truck-multi-drone team logistics
	TSP	Traveling salesman problem
	TSP-D	Traveling salesman problem with drone
	TSP-DS	Traveling salesman problem with a drone station
	TSP-mD	Traveling salesman problem with multi-drone
	2E-VRP	Two-echelon vehicle routing problem
	UAGVS	Unmanned aerial and ground vehicles
	UAVRP	Unmanned aerial vehicles routing problem
	VRP	Vehicle routing problem
	VRPD	Vehicle routing problem with drones
	AAV	Autonomous aerial vehicle
	AV	Air vehicle
	GV	Ground vehicle
	UAV	Unmanned air vehicle
	UGV	Unmanned ground vehicle

Data availability

No data was used for the research described in the article.

References

- [1] J. Reed, Amazon Prime Air is Set to Start Drone Deliveries this Year - Avionics International, 2022. <https://www.avionictoday.com/2022/06/16/amazon-prime-air-drone-deliveries-2022/>.
- [2] F. Tobe, DHL UAS Parcel Delivery Gets First Test - Robohub, 2014. <https://robhub.org/dhl-uas-parcel-delivery-gets-first-test/>.
- [3] J. Daleo, DHL Pulling its Parcelcopter Drone, Ceasing Drone Development, 2021. <https://www.freightwaves.com/news/dhl-pulling-its-parcelcopter-drone-ceasing-drone-development>.
- [4] C. Clover, Alibaba Trials China Drone Deliveries, 2015. <https://www.ft.com/content/3a7f1602-ad0e-11e4-bfcf-00144feab7de>.
- [5] M. Jing, JD.com's Drone Delivery goes into Operation, 2016. https://www.chinadaily.com.cn/business/2016-06/08/content_25655486.htm.
- [6] W. Xu, Meituan's Drone Food Deliveries on the Rise, 2023. <https://www.chinadaily.com.cn/a/202309/19/WS6508e4fca310d2dce4bb66a8.html>.
- [7] J. Deng, J. Hua, B. Adu Gyamfi, R. Shaw, in: Drones Activity in Epidemic Prevention and Prospects in the Post-COVID-19, Springer, Singapore, 2022, pp. 33–43, https://doi.org/10.1007/978-981-16-6959-0_3.
- [8] Y. Choi, P.M. Schonfeld, A comparison of optimized deliveries by drone and truck, Transp. Plan. Technol. 44 (3) (2021) 319–336, <https://doi.org/10.1080/03081060.2021.1883230>.
- [9] I. Khoufi, A. Laouiti, C. Adjih, A survey of recent extended variants of the traveling salesman and vehicle routing problems for unmanned aerial vehicles, Drones-Basel 3 (3) (2019) 66, <https://doi.org/10.3390/drones3030066>.
- [10] G. Macrina, L. Di Puglia Pugliese, F. Guerriero, G. Laporte, Drone-aided routing: a literature review, Transp. Res. Part C Emerg. Technol. 120 (2020) 102762, <https://doi.org/10.1016/j.trc.2020.102762>.
- [11] B. Madani, M. Ndiaye, Hybrid truck-drone delivery systems: a systematic literature review, IEEE Access 10 (2022) 92854–92878, <https://doi.org/10.1109/ACCESS.2022.3202895>.
- [12] J. Chen, X. Zhang, B. Xin, H. Fang, Coordination between unmanned aerial and ground vehicles: a taxonomy and optimization perspective, IEEE T. Cybern. 46 (4) (2016) 959–972, <https://doi.org/10.1109/TCYB.2015.2418337>.
- [13] S.H. Chung, B. Sah, J. Lee, Optimization for drone and drone-truck combined operations: a review of the state of the art and future directions, Comput. Oper. Res. 123 (2020) 105004, <https://doi.org/10.1016/j.cor.2020.105004>.
- [14] A. Otto, N. Agatz, J. Campbell, B. Golden, E. Pesch, Optimization approaches for civil applications of unmanned aerial vehicles (UAVs) or aerial drones: a survey, Networks 72 (4) (2018) 411–458, <https://doi.org/10.1002/net.21818>.
- [15] D.R. Vilorio, E.L. Solano-Charris, A. Munoz-Villamizar, J.R. Montoya-Torres, Unmanned aerial vehicles/drones in vehicle routing problems: a literature review, Int. Trans. Oper. Res. 28 (4) (2021) 1626–1657, <https://doi.org/10.1111/itor.12783>.
- [16] M. Moshref-Javadi, M. Winkenbach, Applications and Research avenues for drone-based models in logistics: a classification and review, Expert Syst. Appl. 177 (2021) 114854, <https://doi.org/10.1016/j.eswa.2021.114854>.
- [17] H. Li, J. Chen, F. Wang, M. Bai, Ground-vehicle and unmanned-aerial-vehicle routing problems from two-echelon scheme perspective: a review, Eur. J. Oper. Res. 294 (3) (2021) 1078–1095, <https://doi.org/10.1016/j.ejor.2021.02.022>.
- [18] N. Sluijk, A.M. Florio, J. Kinable, N. Dellaert, T. Van Woensel, Two-echelon vehicle routing problems: a literature review, Eur. J. Oper. Res. 304 (3) (2023) 865–886, <https://doi.org/10.1016/j.ejor.2022.02.022>.
- [19] M. Boccia, A. Mancuso, A. Masone, T. Murino, C. Sterle, New features for customer classification in the flying sidekick traveling salesman problem, Expert Syst. Appl. 247 (2024) 123106, <https://doi.org/10.1016/j.eswa.2023.123106>.
- [20] S. Poikonen, B. Golden, E.A. Wasil, A branch-and-bound approach to the traveling salesman problem with a drone, INFORMS J. Comput. 31 (2) (2019) 335–346, <https://doi.org/10.1287/ijoc.2018.0826>.
- [21] S. Poikonen, B. Golden, The mothership and drone routing problem, INFORMS J. Comput. 32 (2) (2020) 249–262, <https://doi.org/10.1287/ijoc.2018.0879>.
- [22] M. Dell'Amico, R. Montemanni, S. Novellani, Algorithms based on branch and bound for the flying sidekick traveling salesman problem, Omega-Int. J. Manage. Sci. 104 (2021) 102493, <https://doi.org/10.1016/j.omega.2021.102493>.
- [23] E. Teimoury, R. Rashid, The sustainable hybrid truck-drone delivery model with stochastic customer existence, Res. Transp. Econ. 100 (2023), <https://doi.org/10.1016/j.retrec.2023.101325>.
- [24] M. Dell'Amico, R. Montemanni, S. Novellani, Drone-assisted deliveries: new formulations for the flying sidekick traveling salesman problem, Optim. Lett. 15 (5) (2021) 1617–1648, <https://doi.org/10.1007/s11590-019-01492-z>.
- [25] M. Dell'Amico, R. Montemanni, S. Novellani, Exact models for the flying sidekick traveling salesman problem, Int. Trans. Oper. Res. 29 (3) (2022) 1360–1393, <https://doi.org/10.1111/itor.13030>.
- [26] D. Schermer, M. Moeini, O. Wendt, A branch-and-cut approach and alternative formulations for the traveling salesman problem with drone, Networks 76 (2) (2020) 164–186, <https://doi.org/10.1002/net.21958>.
- [27] M. Dell'Amico, R. Montemanni, S. Novellani, Modeling the flying sidekick traveling salesman problem with multiple drones, Networks 78 (3) (2021) 303–327, <https://doi.org/10.1002/net.22022>.
- [28] S.G. Manyam, D.W. Casbeer, K. Sundar, Path Planning for Cooperative Routing of Air-Ground Vehicles, American Control Conference (ACC) (2016) 4630–4635, [doi: 10.1109/ACC.2016.7526082](https://doi.org/10.1109/ACC.2016.7526082).
- [29] S.G. Manyam, K. Sundar, D.W. Casbeer, Cooperative routing for an air-ground vehicle team-exact algorithm, transformation method, and heuristics, IEEE Trans. Autom. Sci. Eng. 17 (1) (2020) 537–547, <https://doi.org/10.1109/TASE.2019.2931894>.

- [194] H. Baik, J. Valenzuela, An optimization drone routing model for inspecting wind farms, *Soft Comput.* 25 (3) (2021) 2483–2498, <https://doi.org/10.1007/s00500-020-05316-6>.
- [195] L. Amorosi, J. Puerto, C. Valverde, Coordinating drones with mothership vehicles: the mothership and drone routing problem with graphs, *Comput. Oper. Res.* 136 (2021) 105445, <https://doi.org/10.1016/j.cor.2021.105445>.
- [196] C.C. Chinadaily, Drone Used to help Sea Patrol in Zhanjiang, 2016. http://www.chinadaily.com.cn/m/guangdong/zhanjiang/2016-05/13/content_25259420.htm.
- [197] Xinhua, Drones Used to Monitor Traffic during May Day Holiday in S China [1]-Chinadaily.com.cn (2016). http://www.chinadaily.com.cn/china/2016-05/03/content_25017530.htm.
- [198] J. Li, G. Deng, C. Luo, Q. Lin, Q. Yan, Z. Ming, A hybrid path planning method in unmanned air/ground vehicle (UAV/UGV) cooperative systems, *IEEE Trans. Veh. Technol.* 65 (12) (2016) 9585–9596, <https://doi.org/10.1109/TVT.2016.2623666>.
- [199] J. Qiu, D. Grace, G. Ding, M.D. Zakaria, Q. Wu, Air-ground heterogeneous networks for 5G and beyond via integrating high and low altitude platforms, *IEEE Wirel. Commun.* 26 (6) (2019) 140–148, <https://doi.org/10.1109/MWC.0001.1800575>.
- [200] W. Wang, Y. Liu, R. Srikant, L. Ying, 3M-RL: multi-resolution, multi-agent, mean-field reinforcement learning for autonomous UAV routing, *IEEE Trans. Intell. Transp. Syst.* 23 (7) (2022) 8985–8996, <https://doi.org/10.1109/TITS.2021.3089120>.
- [201] Y. Yu, J. Guo, M. Chadli, Z. Xiang, Distributed adaptive fuzzy formation control of uncertain multiple unmanned aerial vehicles with actuator faults and switching topologies, *IEEE Trans. Fuzzy Syst.* 31 (3) (2023) 919–929, <https://doi.org/10.1109/TFUZZ.2022.3193440>.
- [202] A. Heidari, N.J. Navimipour, M. Unal, G. Zhang, Machine learning applications in internet-of-drones: systematic review, recent deployments, and open issues, *ACM Comput. Surv.* 55 (12) (2023) 1–45, <https://doi.org/10.1145/3571728>.